

Act IV: The Play Begins

Market Equilibrium, Price Setting, Taxes

Module X: Fair Play in the Town Square

Efficiency, Price Setting and Taxes



Previously in Marginal Falls

- ◆ Buyers gave us Demand.
- ◆ Sellers gave us Supply.
- ◆ Buyers and Sellers interact in the Market.
- ◆ They agree at the Market Equilibrium.

Next: What value does this interaction generates for the economy?

The Berry Vault

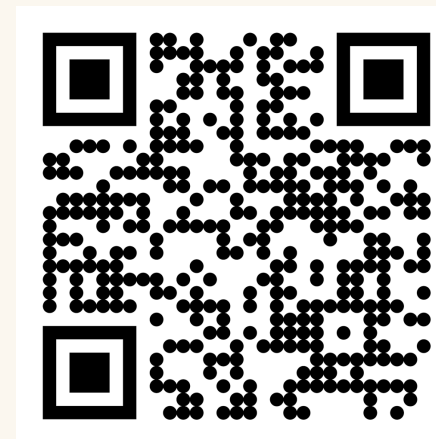
econgames.promitkchaudhuri.com/surplus

Founders' Night pastries need:

1 chocolate filling + 1 berry

Complete pastries: +100

Incomplete pastries: -40



Demand for Berries

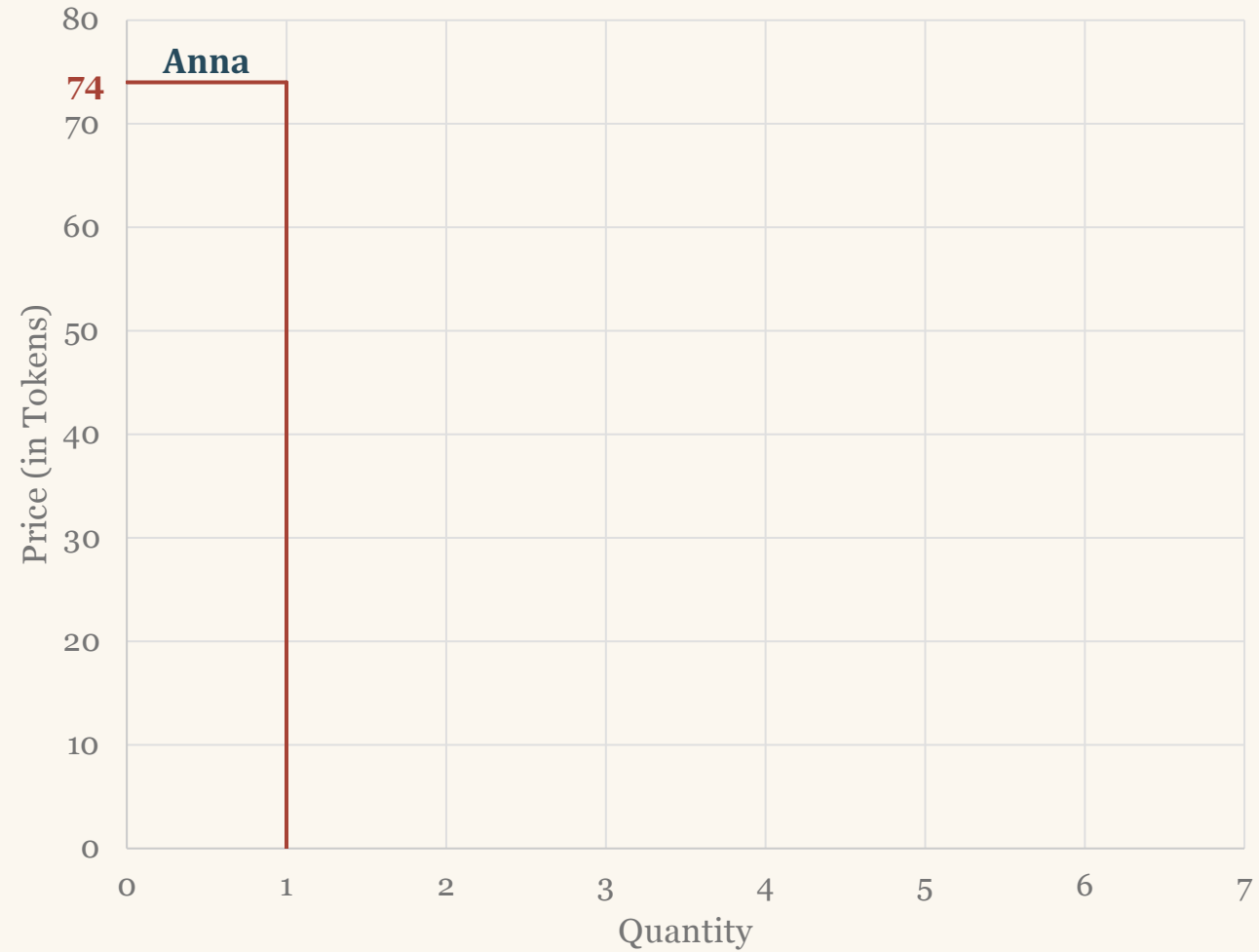
Citizens	Price
Anna	74
Matt	54
Chris	34
Ben	21
Lima	17
Takashi	10
Rima	3

Willingness to Pay:

The maximum price a consumer is willing to pay for a good or service.

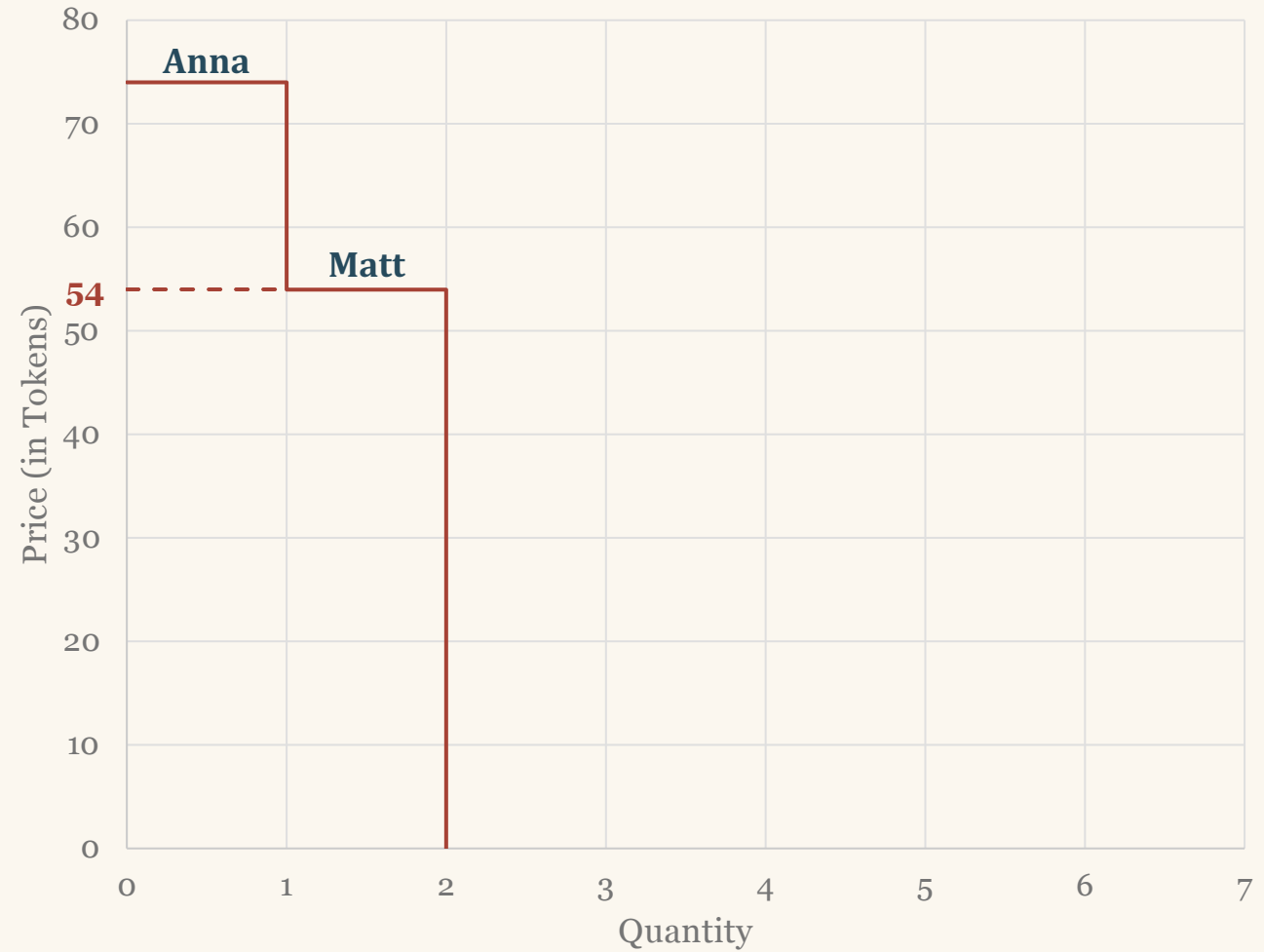
Demand for Berries

Citizens	Max Price
Anna	74
Matt	54
Chris	34
Ben	21
Lima	17
Takashi	10
Rima	3



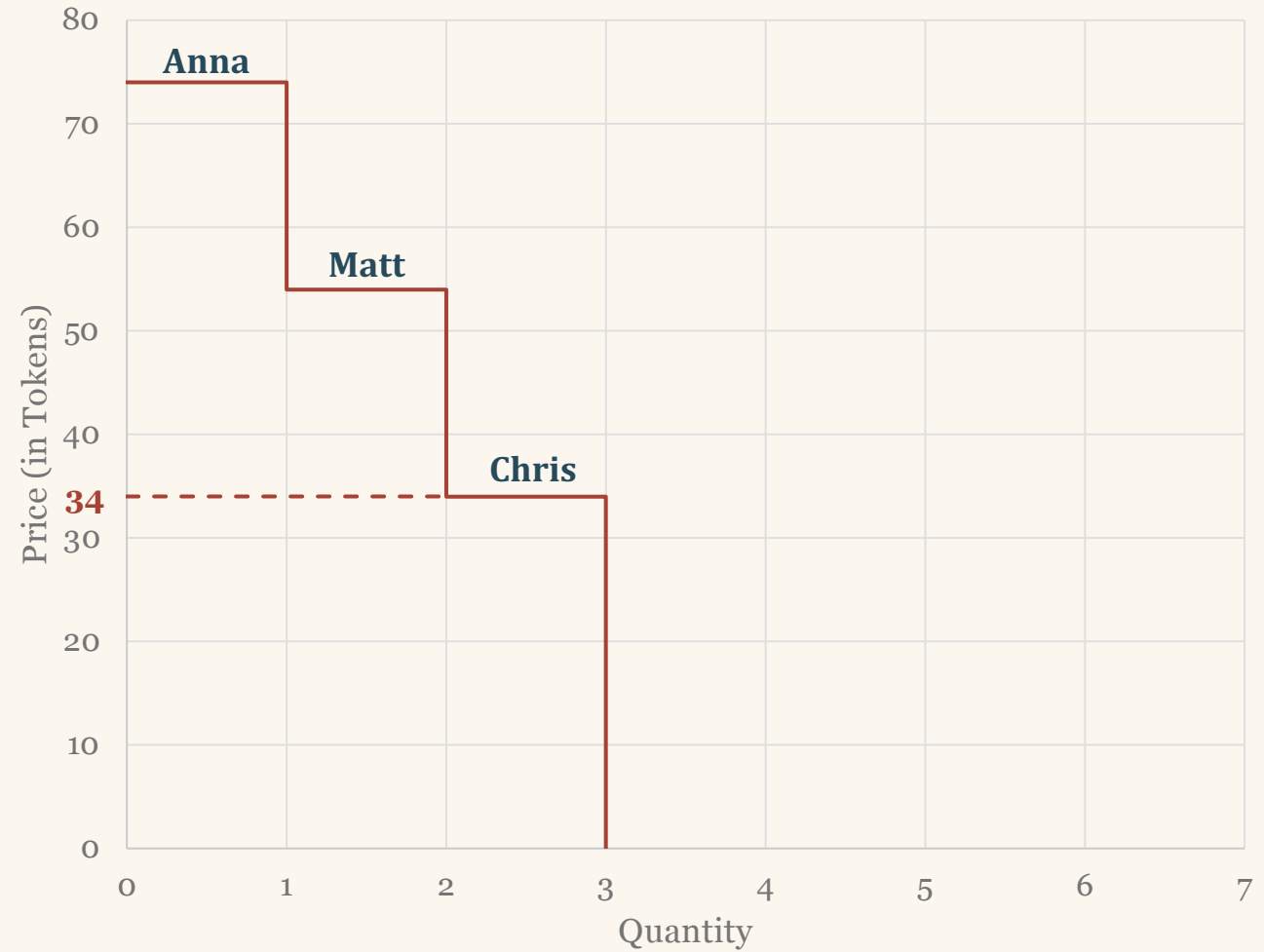
Demand for Berries

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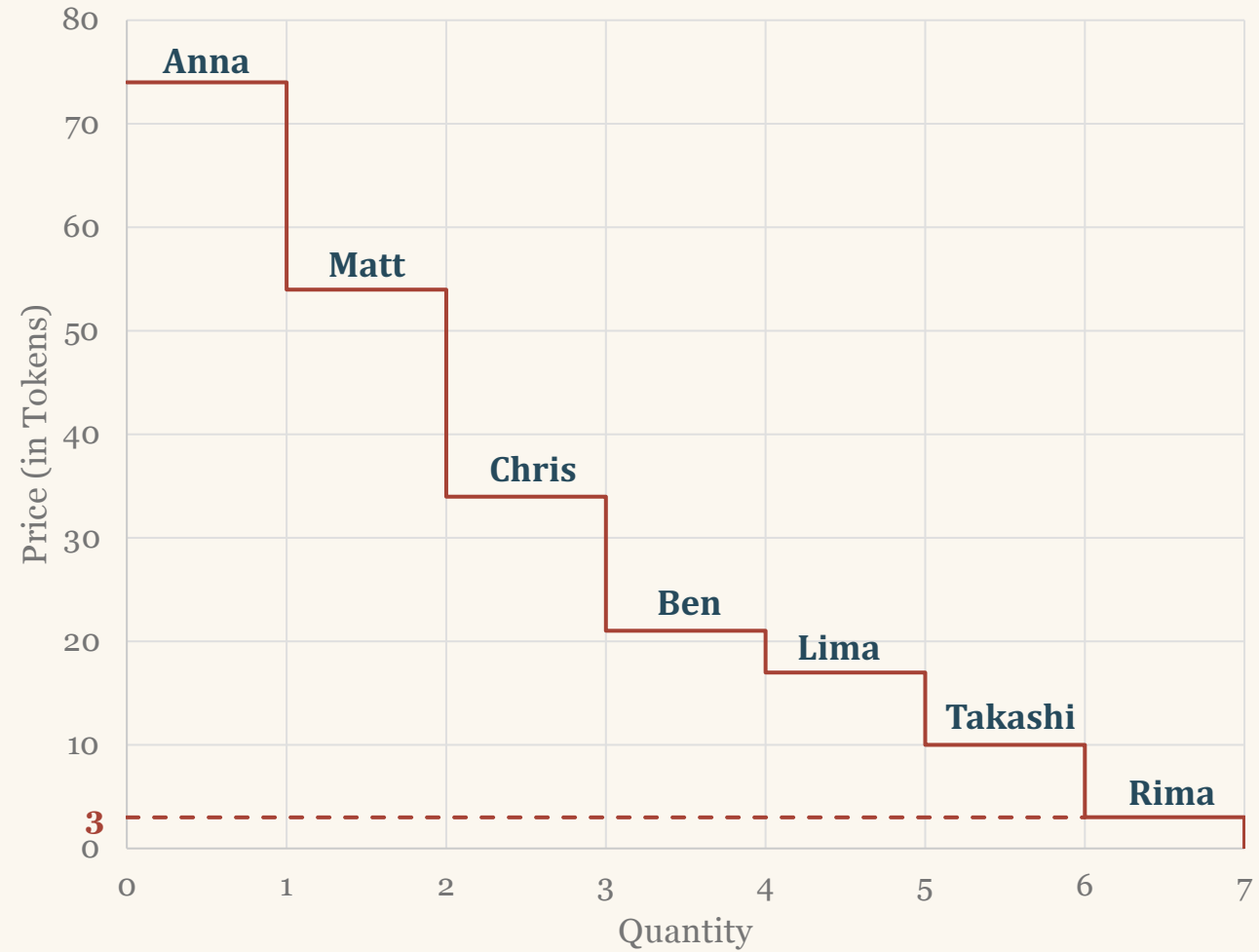
Demand for Berries

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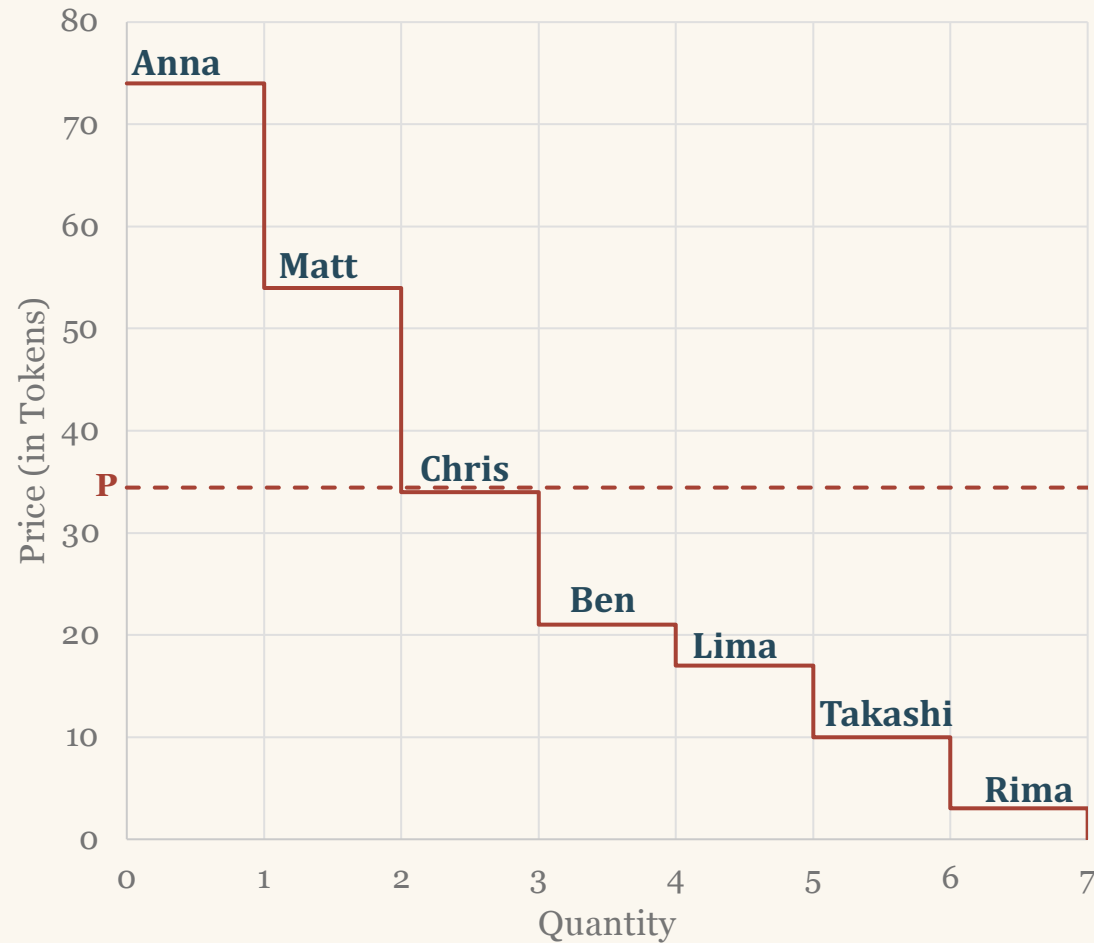


Demand for Berries

Citizens	Max Price
Anna	74
Matt	54
Chris	34
Ben	21
Lima	17
Takashi	10
Rima	3



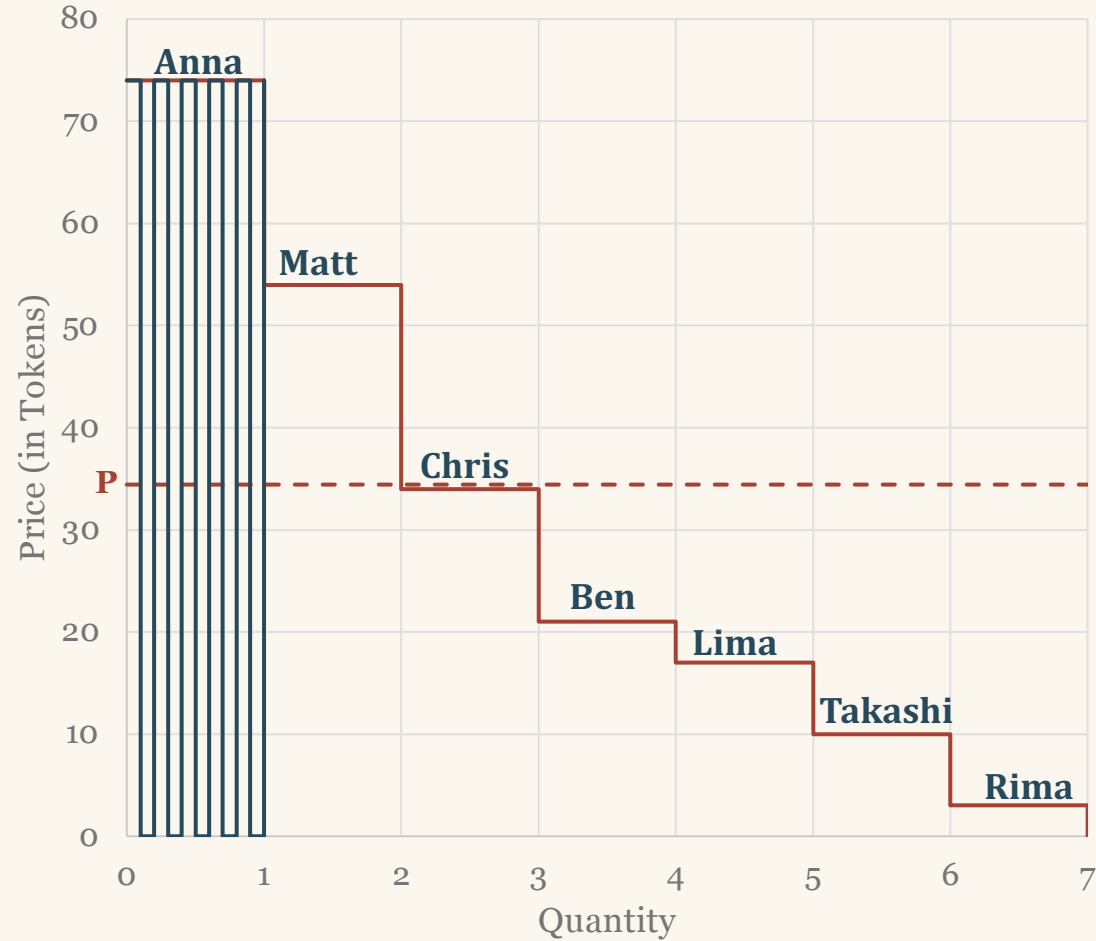
Bank Price



Bank Price (P) = 34.43 Tokens

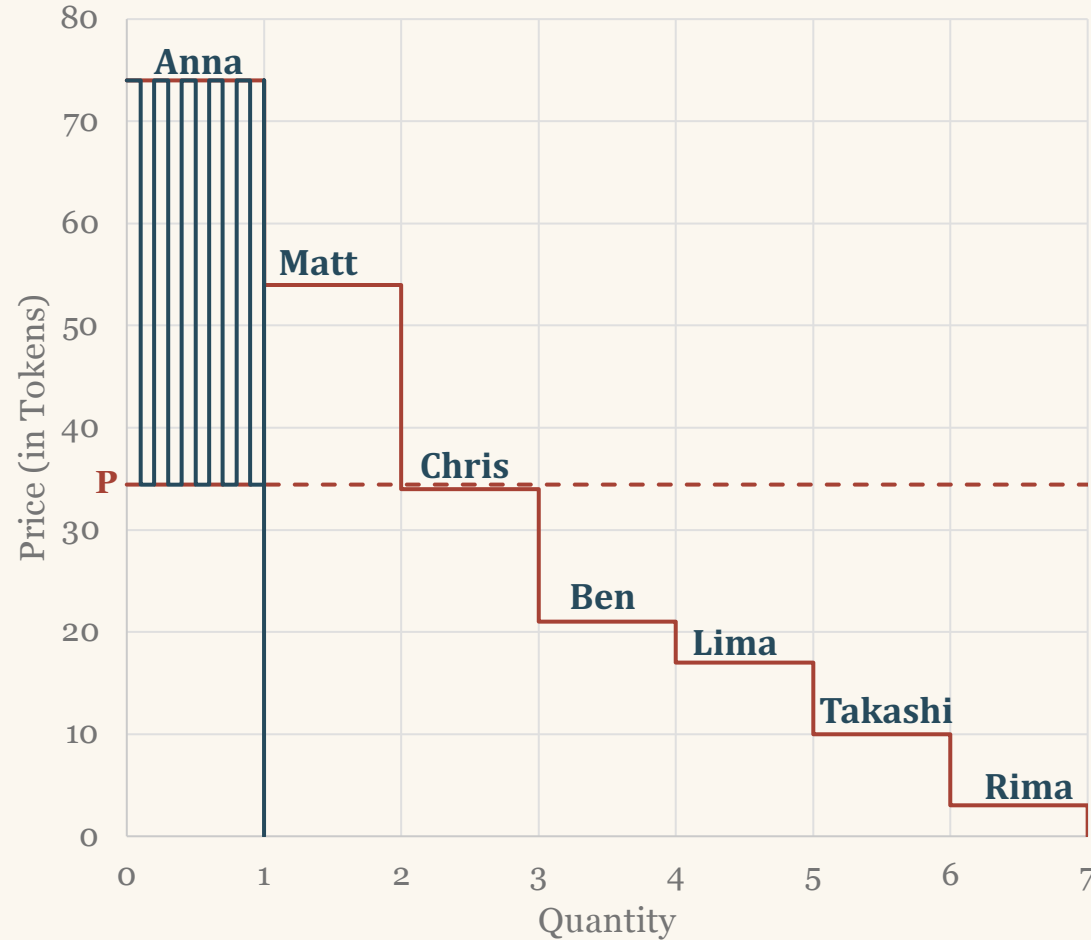
- Anna, Matt > 34.43 (Buy)
- Chris, Ben, Lima, Takashi, Rima < 34.43 (Can't Buy)

Consumer Surplus



Anna:
Valued at 74 Tokens

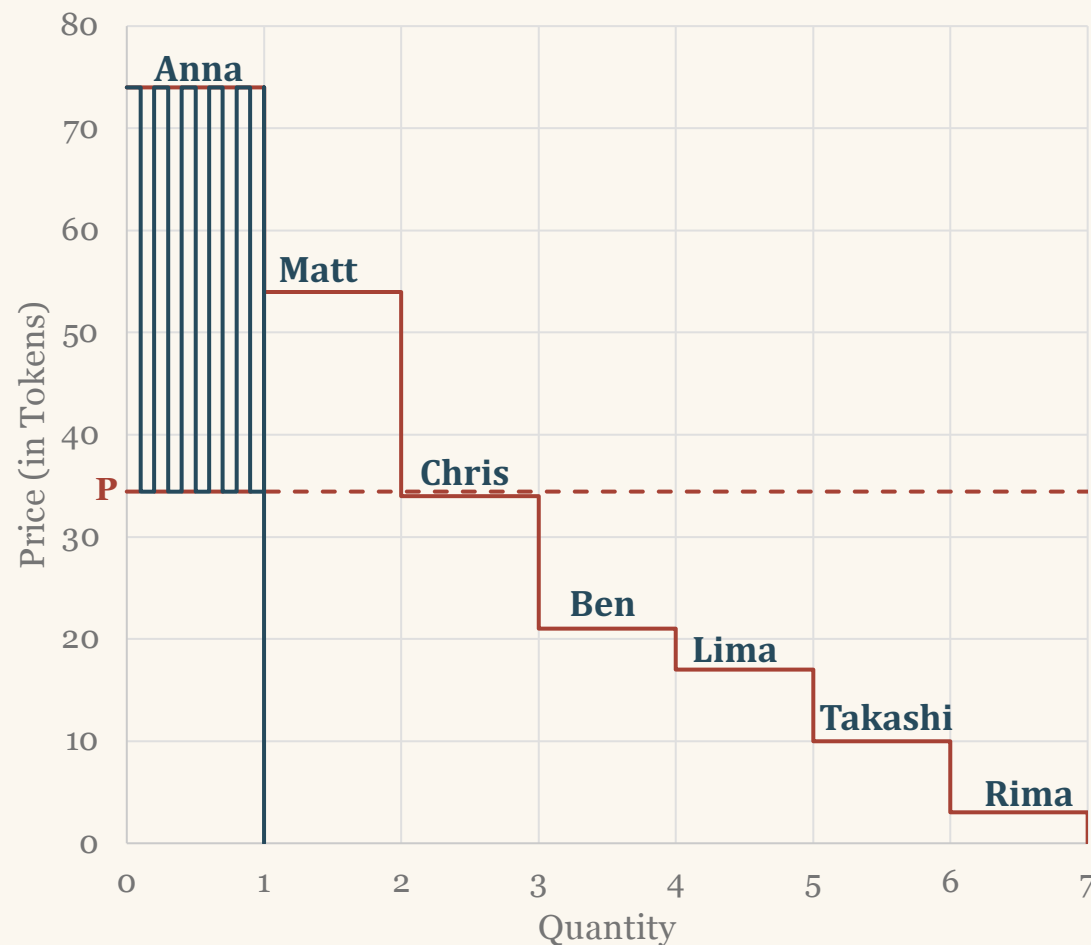
Consumer Surplus



Anna:
Valued at 74 Tokens
Paid 34.43 Tokens

Anna's gain from Trading:
 $74 - 34.43 = 39.57$

Consumer Surplus

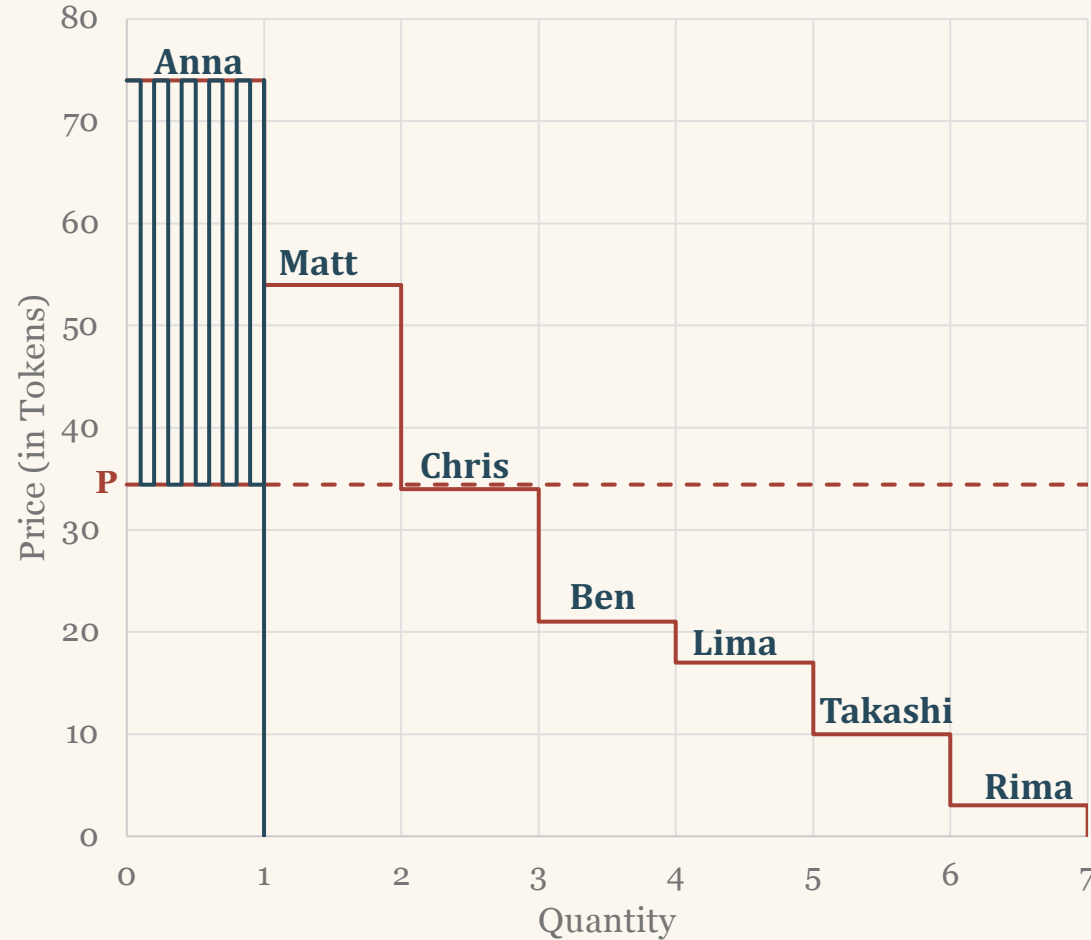


Consumer Surplus:

The difference between the **highest price a consumer is willing to pay** for a good or service and **the actual price the consumer pays**.

Net Benefit of the consumers by participating in the market.

Consumer Surplus

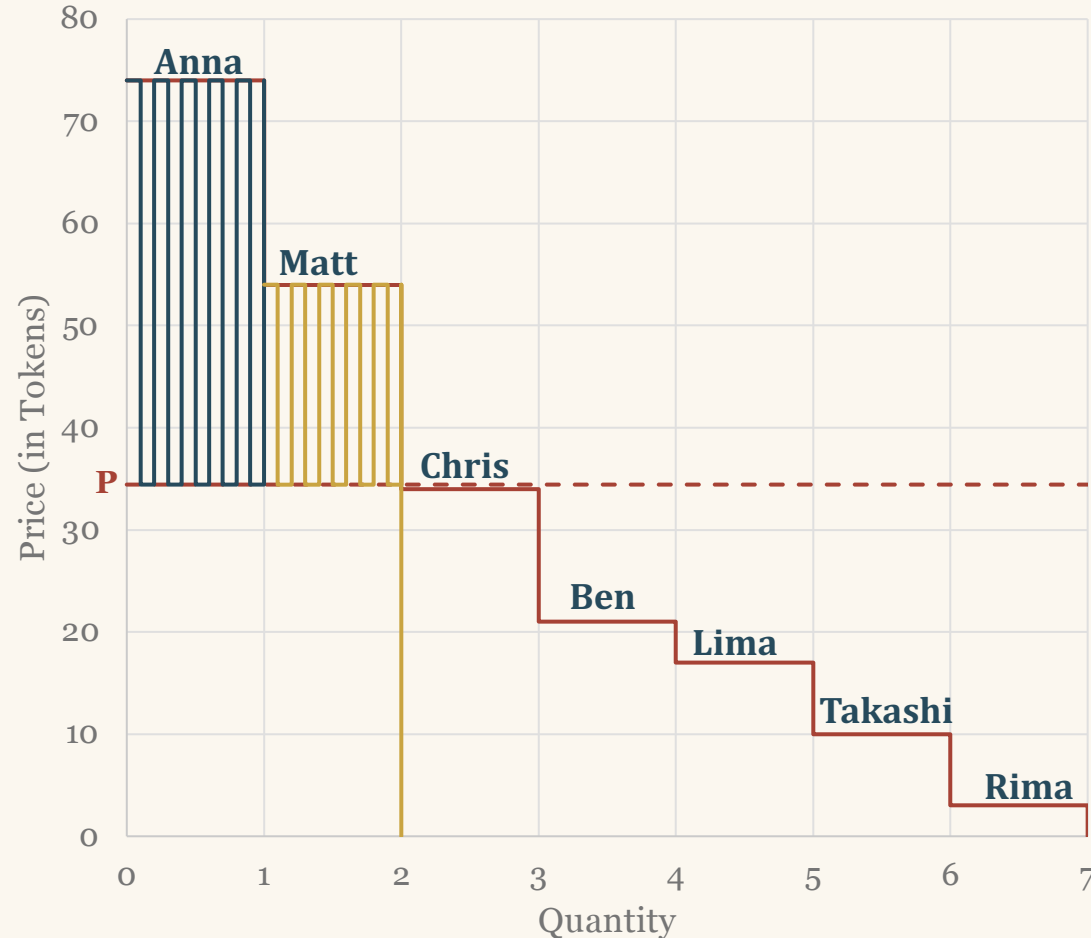


Consumer Surplus:

The difference between the **highest price a consumer is willing to pay** for a good or service and **the actual price the consumer pays**.

Max Price – Bank's Price

Consumer Surplus in the Berry Market

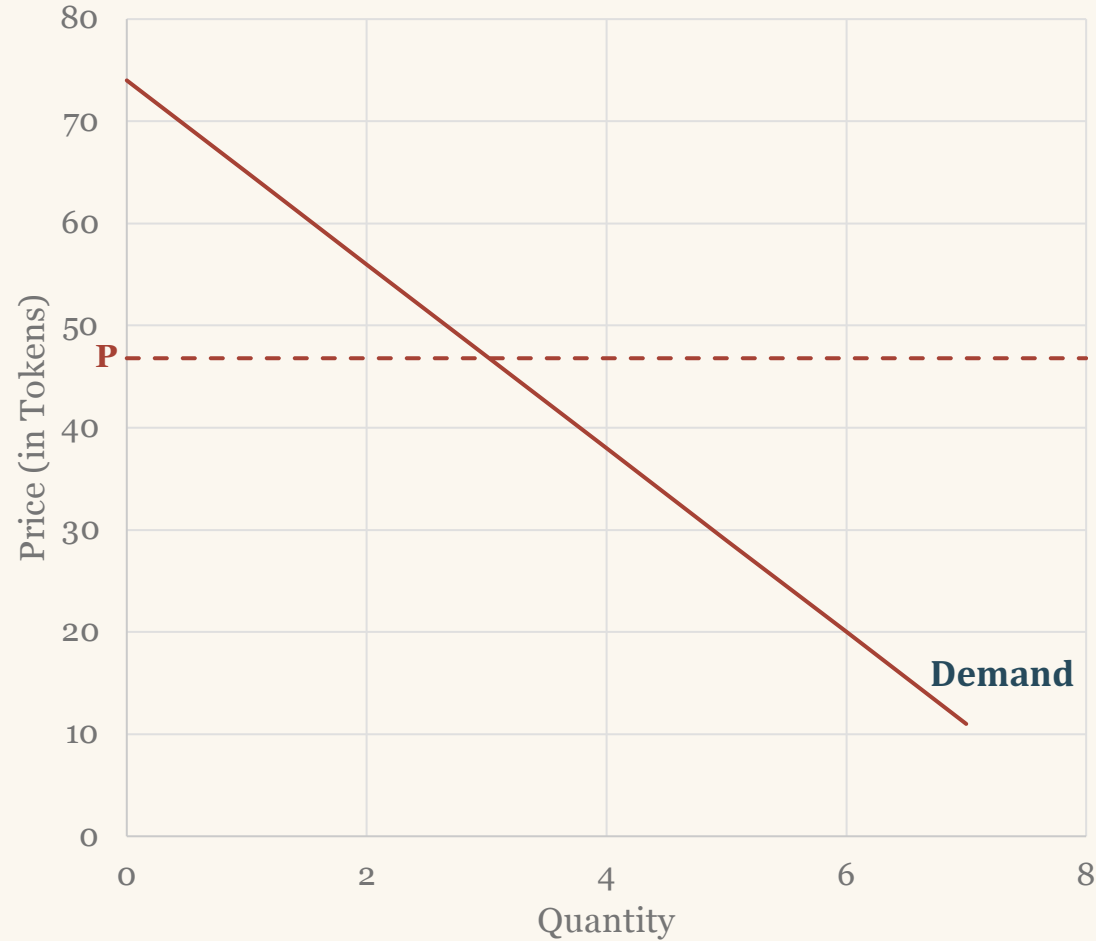


Consumer Surplus in the berry market:

Sum of the areas in blue and yellow.

Sum of the Consumer Surplus of all the consumers in the economy.

Total Consumer Surplus

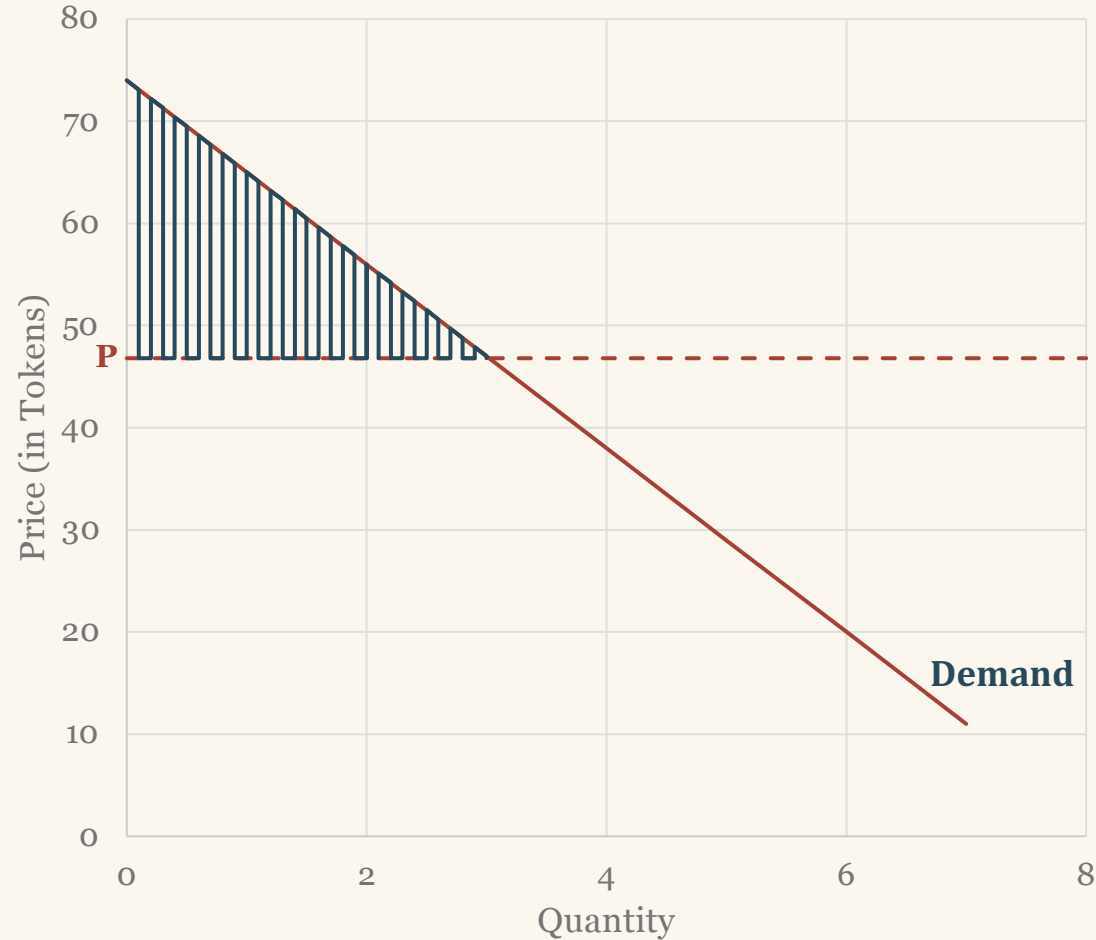


At the Bank price:

Buyers above the price gain surplus.

Consumer Surplus is the area below Demand and above Price.

Total Consumer Surplus



At the Bank price:
Buyers above the price gain surplus.

Consumer Surplus is the area below
Demand and above Price.

Supply for Berries

Citizens	Min Price
Anna	23
Matt	45
Chris	55
Ben	62
Lima	68
Takashi	77
Rima	79

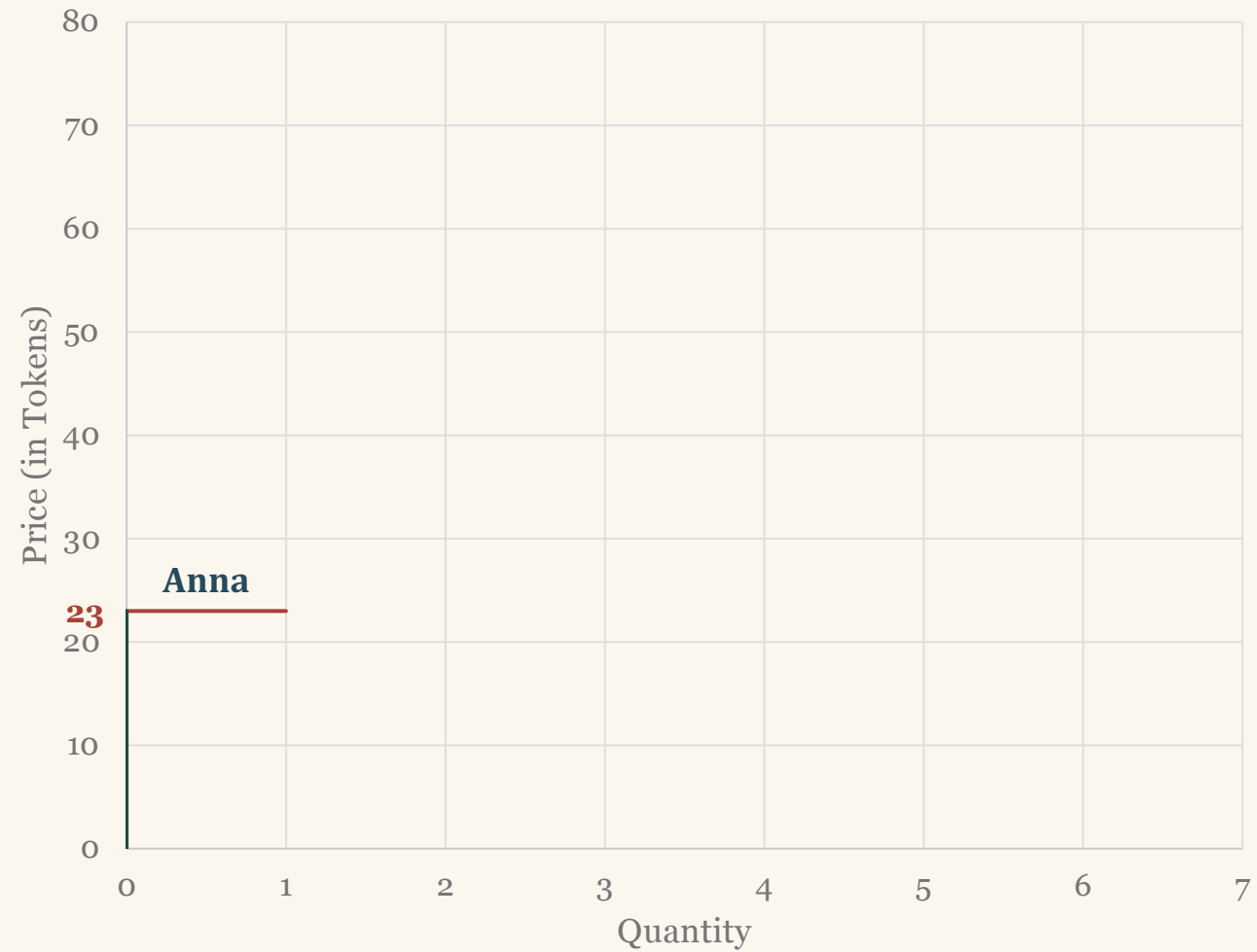
Marginal Cost:

The additional cost to a firm of producing one more unit of a good or service.

The Producers don't want to sell any product below the Marginal Cost.

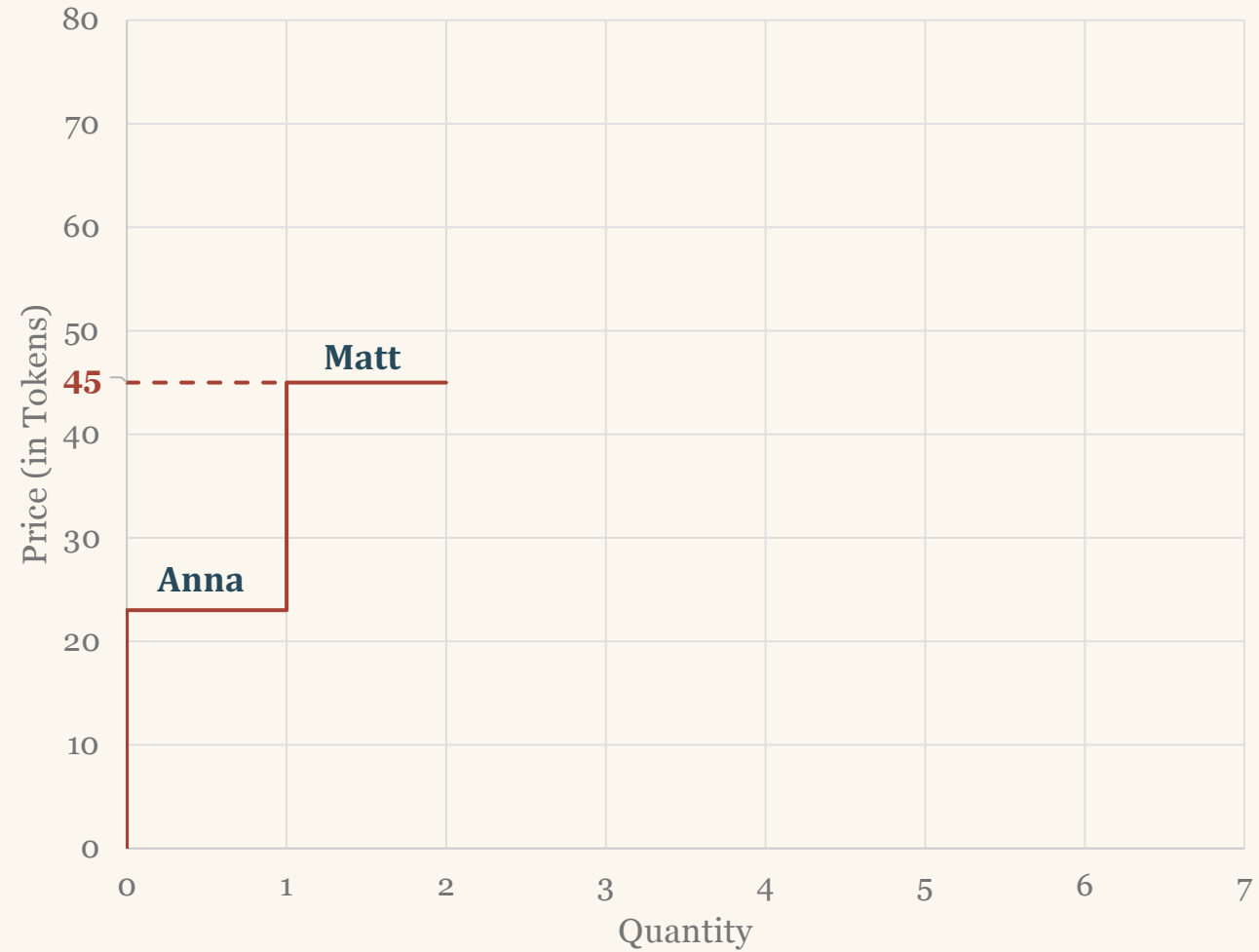
Supply for Berries

Citizens	Min Price
Anna	23
Matt	45
Chris	55
Ben	62
Lima	68
Takashi	77
Rima	79



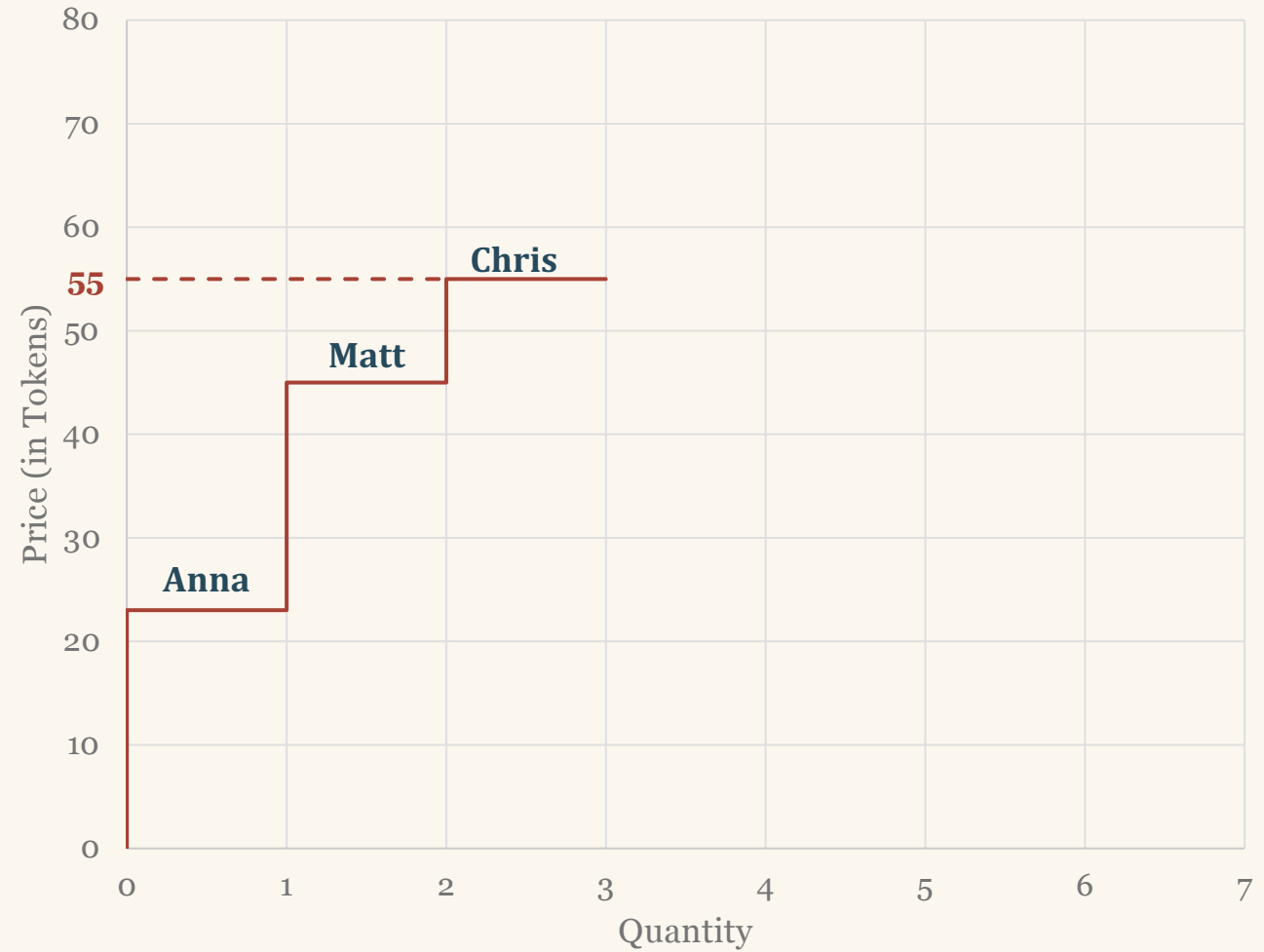
Supply for Berries

Citizens	Min Price
Anna	23
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Ben	62
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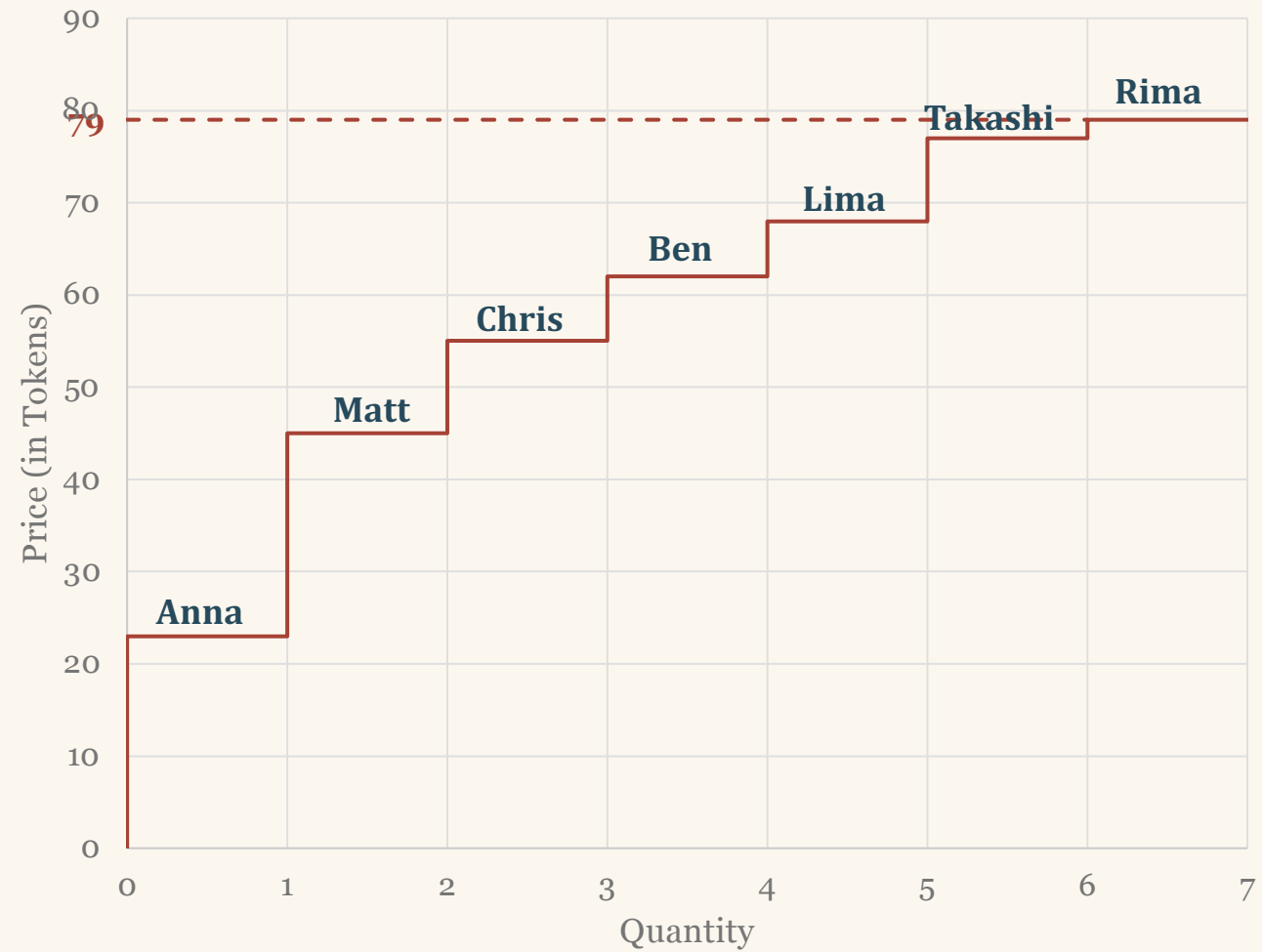
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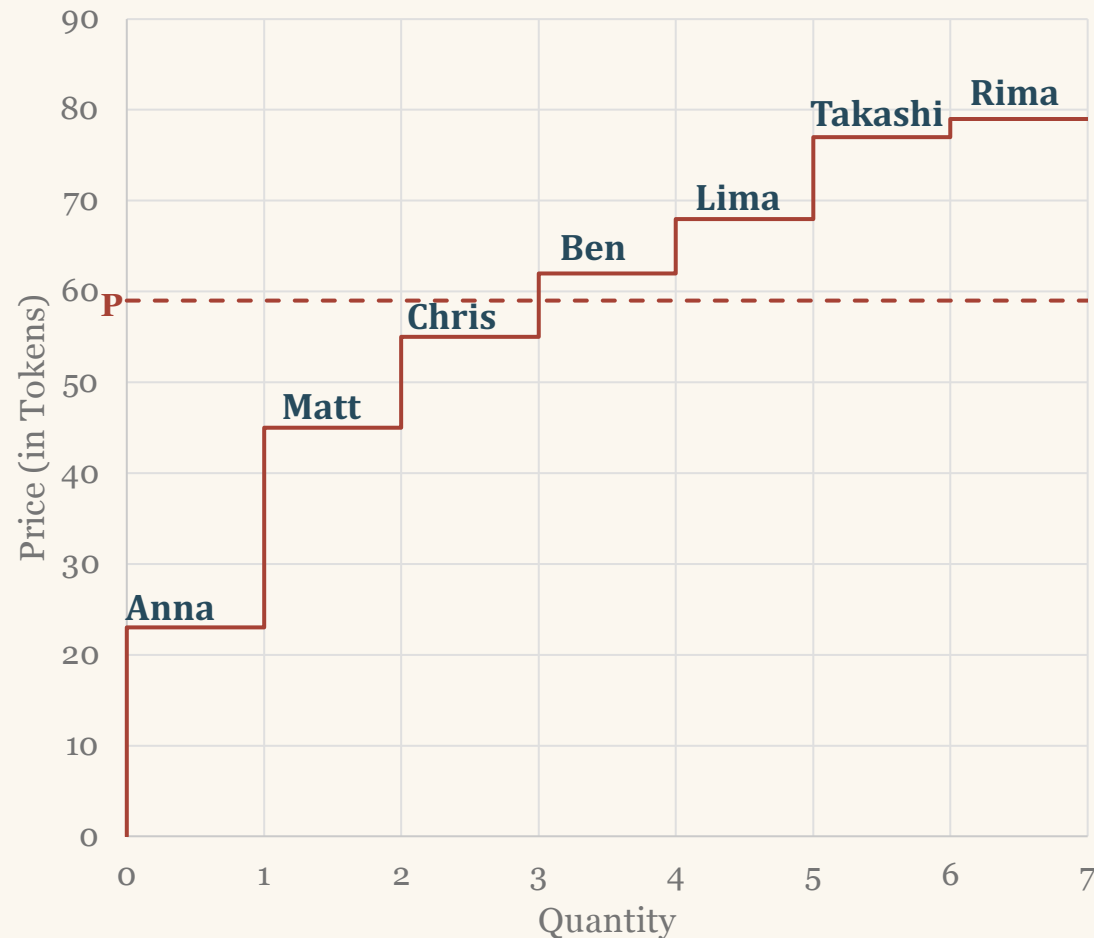


Supply for Berries

Citizens	Min Price
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Matt	45
Chris	55
Ben	62
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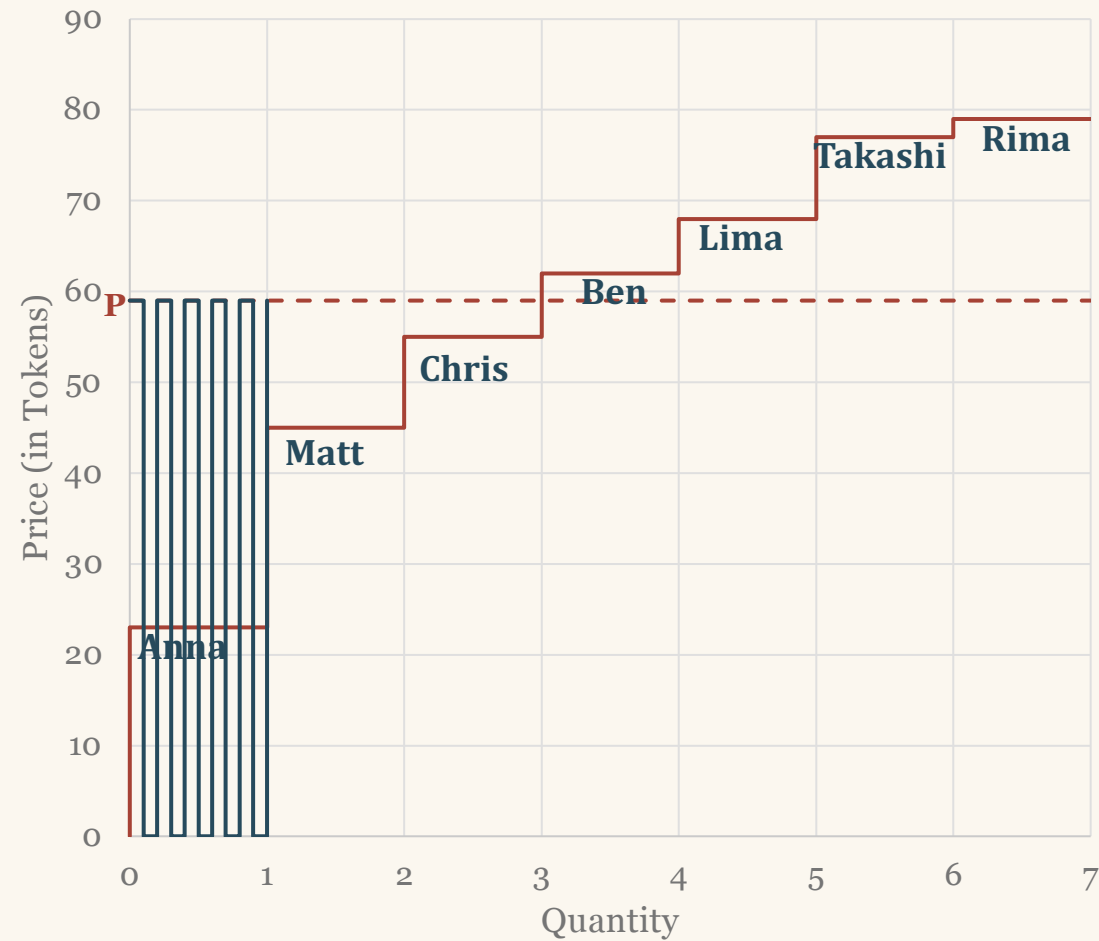
Bank Price



Bank Price (P) = 59 Tokens

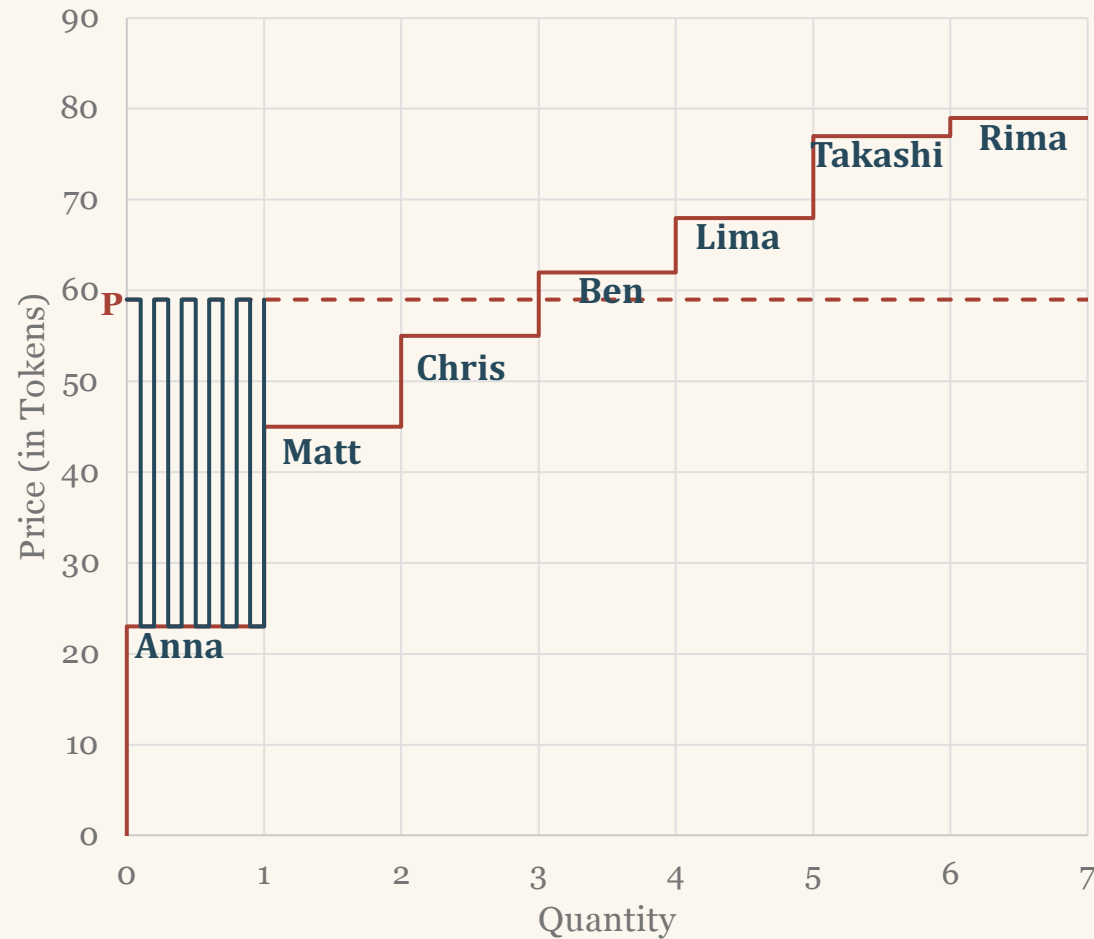
- Anna, Matt, Chris < 59 (Sell)
- Ben, Lima, Takashi, Rima > 59 (Can't Sell)

Bank Price



Anna:
Profit of 59 Tokens

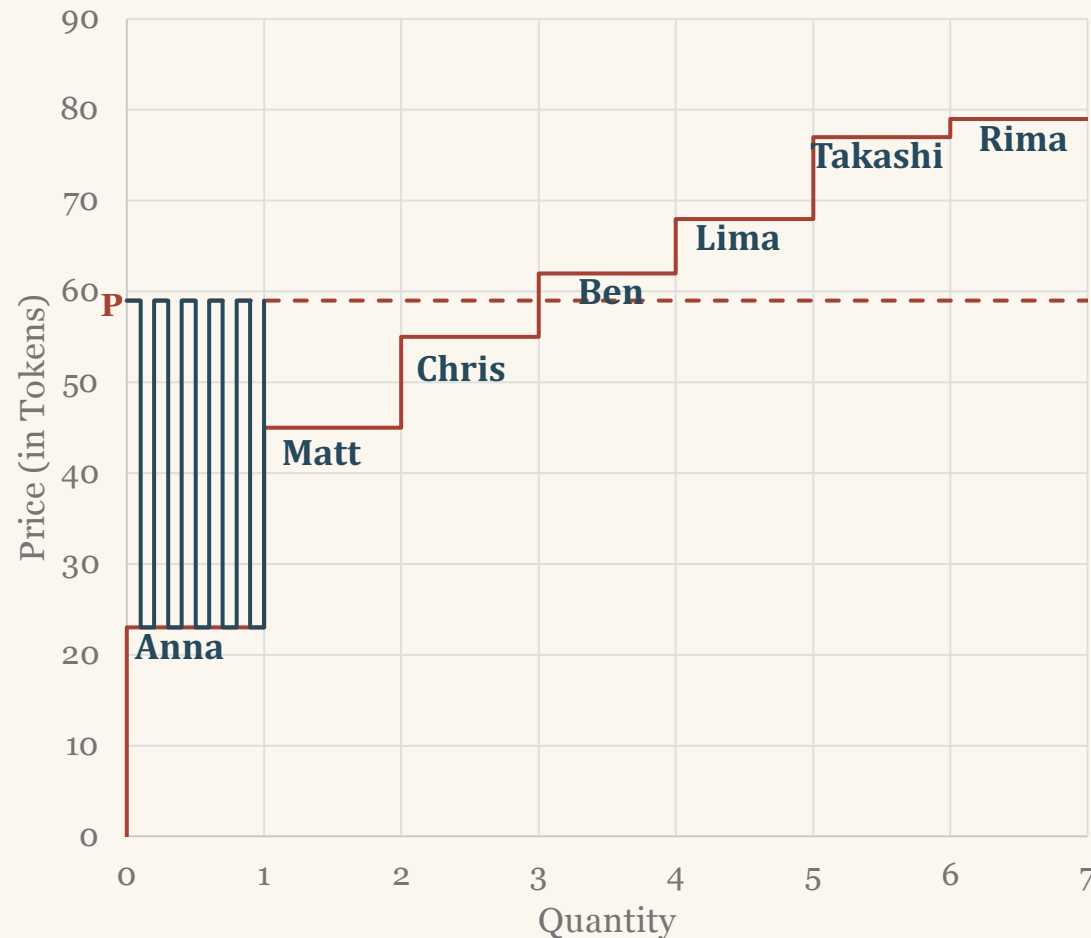
Bank Price



Anna:
Profit of 59 Tokens
Marginal Cost of 23 Tokens.

Anna's gain from Trading:
 $59 - 23 = 36$ Tokens

Producer Surplus

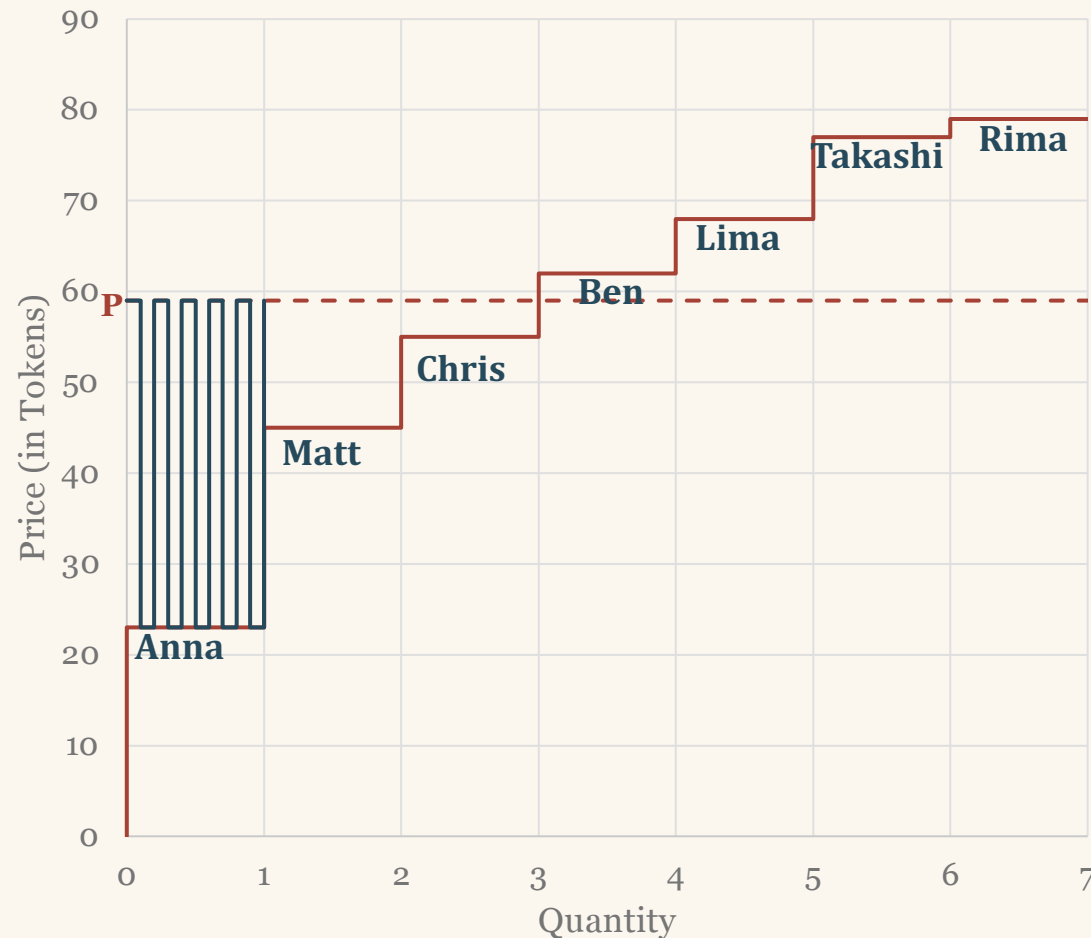


Producer Surplus:

The difference between the **lowest price a firm would be willing to accept** for a good or service and **the price it actually receives**.

Net Benefit of the producers by participating in the market.

Producer Surplus

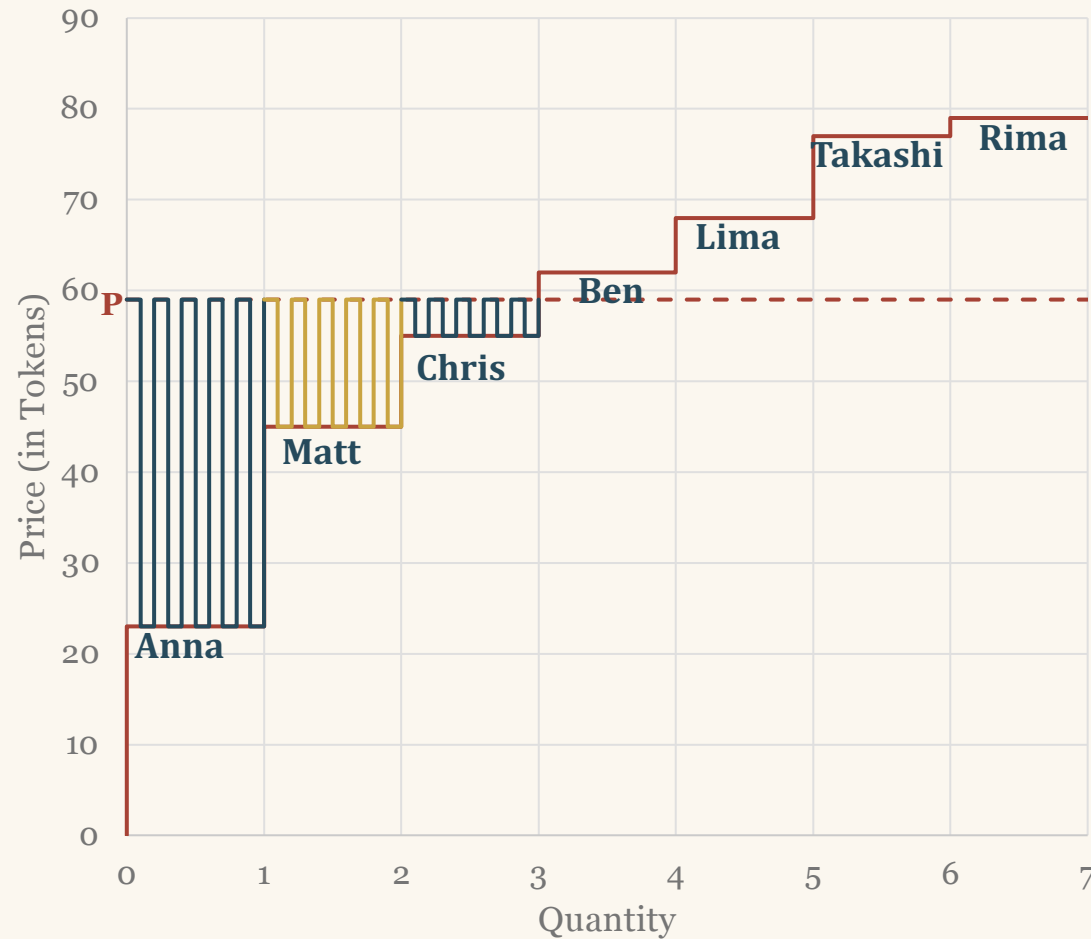


Producer Surplus:

The difference between the **lowest price a firm would be willing to accept** for a good or service and **the price it actually receives**.

Bank's Price – Min Price

Producer Surplus in the Berry Market

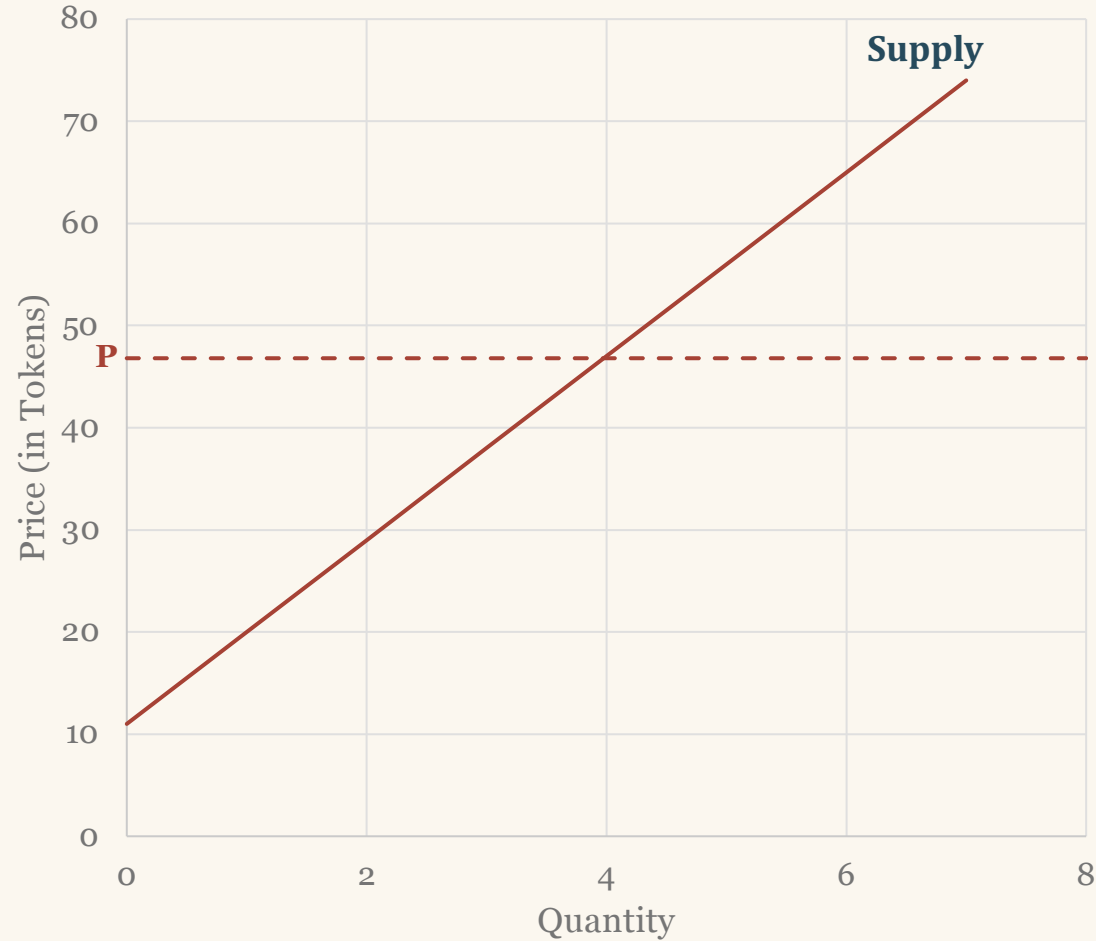


Producer Surplus in the berry market:

Sum of the areas in blue, green and yellow.

Sum of the Producer Surplus of all the consumers in the economy.

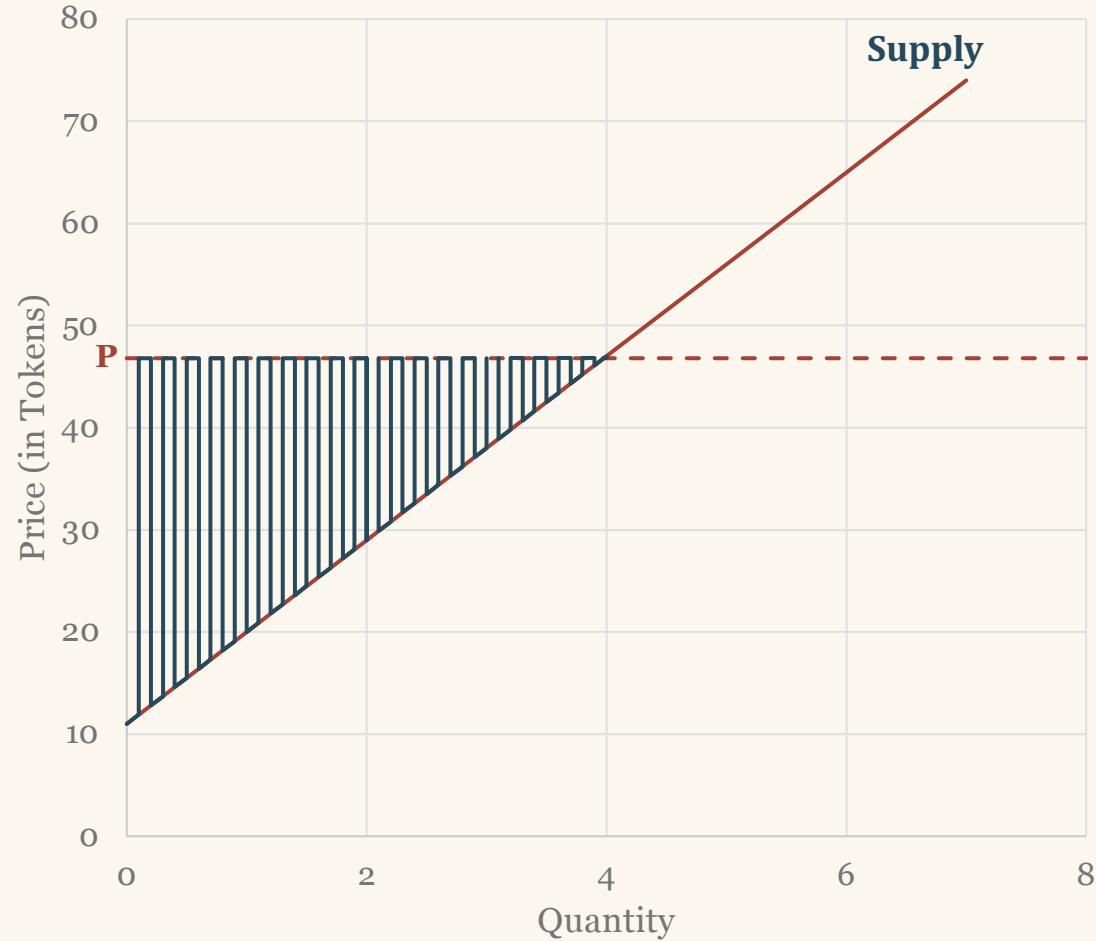
Total Producer Surplus



At the Bank price:
Sellers below the price gain surplus.

Producer Surplus is the area below
Price and above Supply.

Total Producer Surplus



At the Bank price:
Sellers below the price gain surplus.

Producer Surplus is the area below
Price and above Supply.

Efficiency

Total Surplus:

Consumer Surplus + Producer Surplus

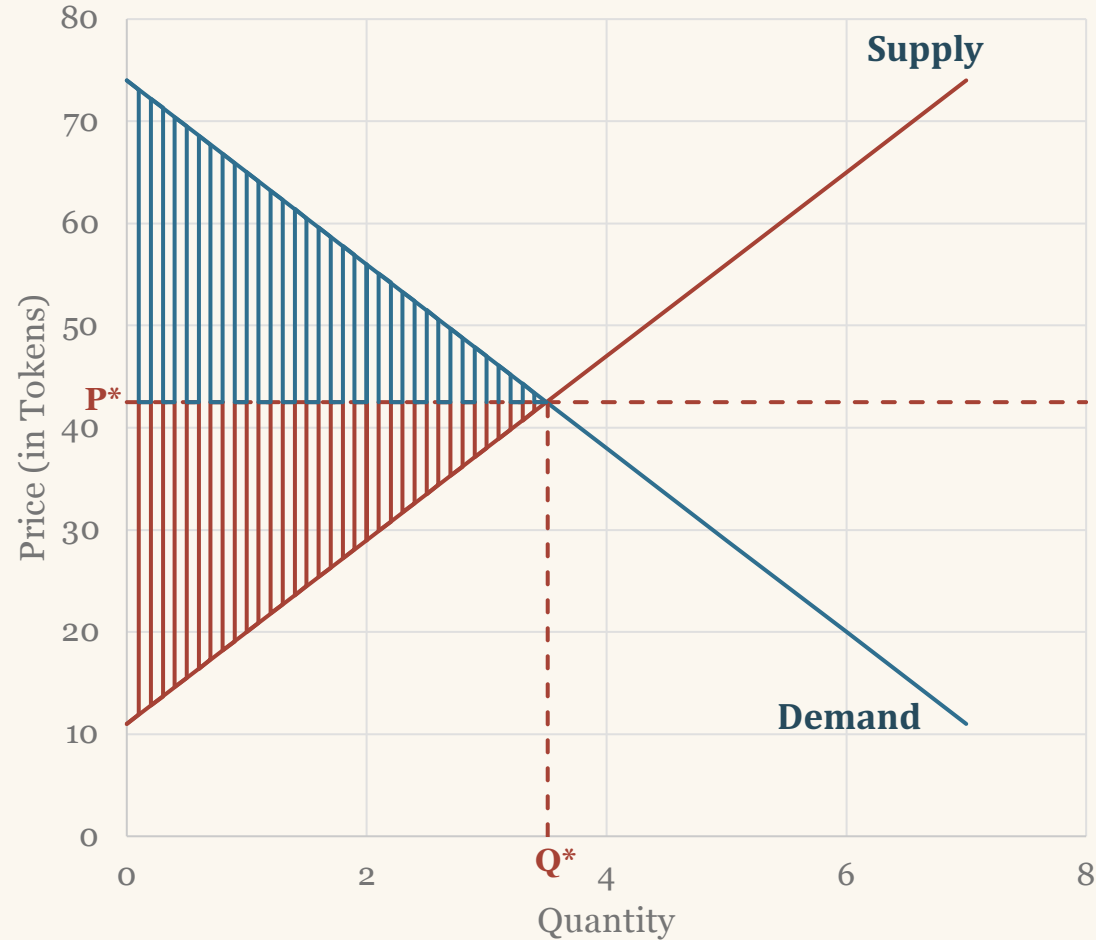
Efficiency:

- A market is efficient if all trades take place where $MB = MC$, and no other trades take place.

OR

- A market is efficient if it maximizes the total surplus, known as the Economic Surplus.

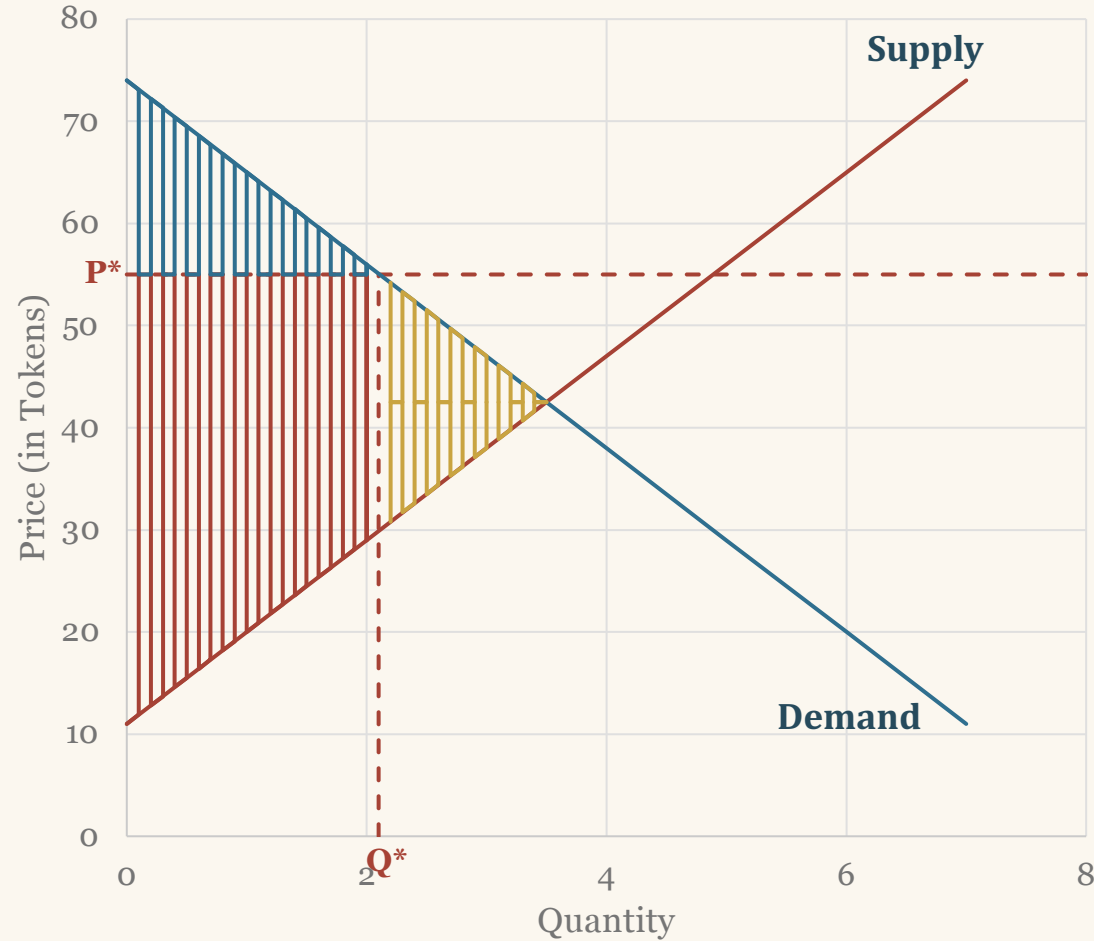
Market Equilibrium



$MB=MC$
Only at Market Equilibrium.

Economic Surplus is also maximized here.

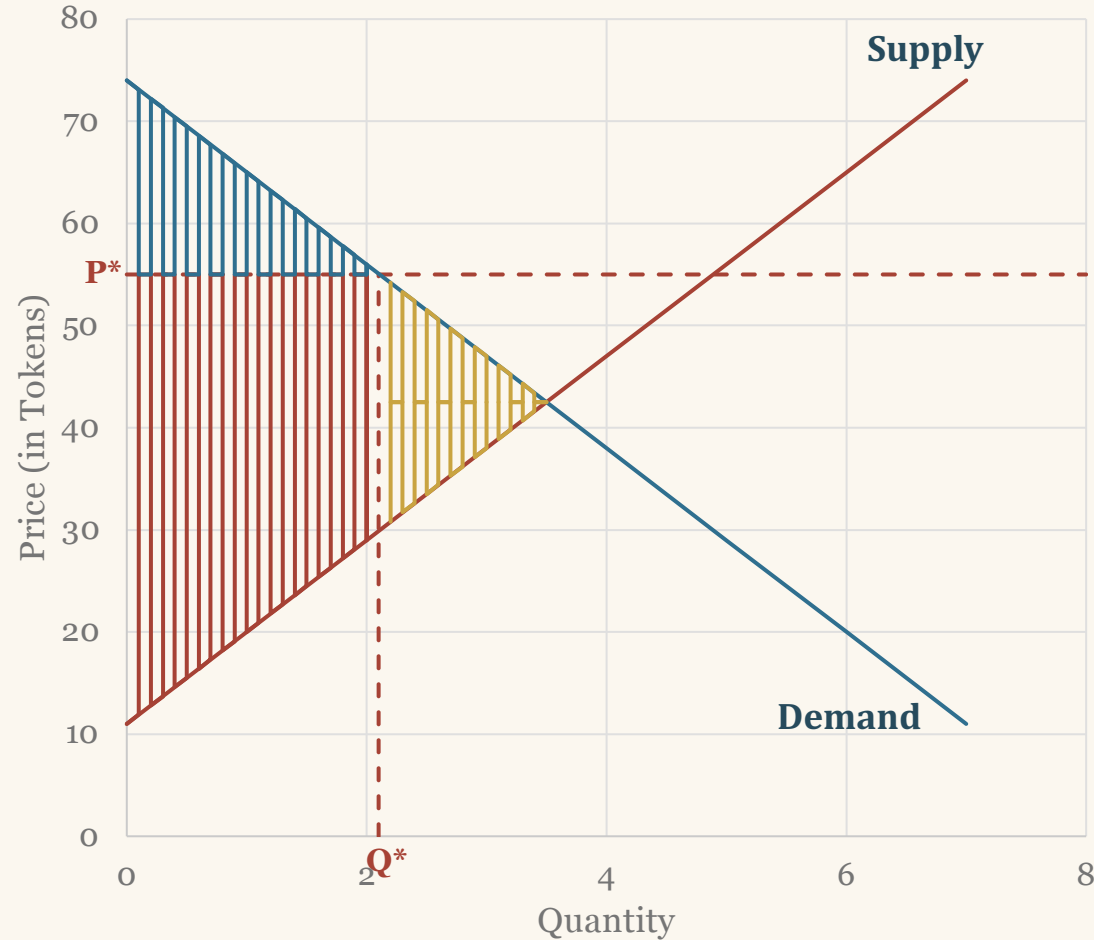
Mayor Intervenes



Mayor ignores the market and increases the price to P^* .

A **yellow** region gets generated.

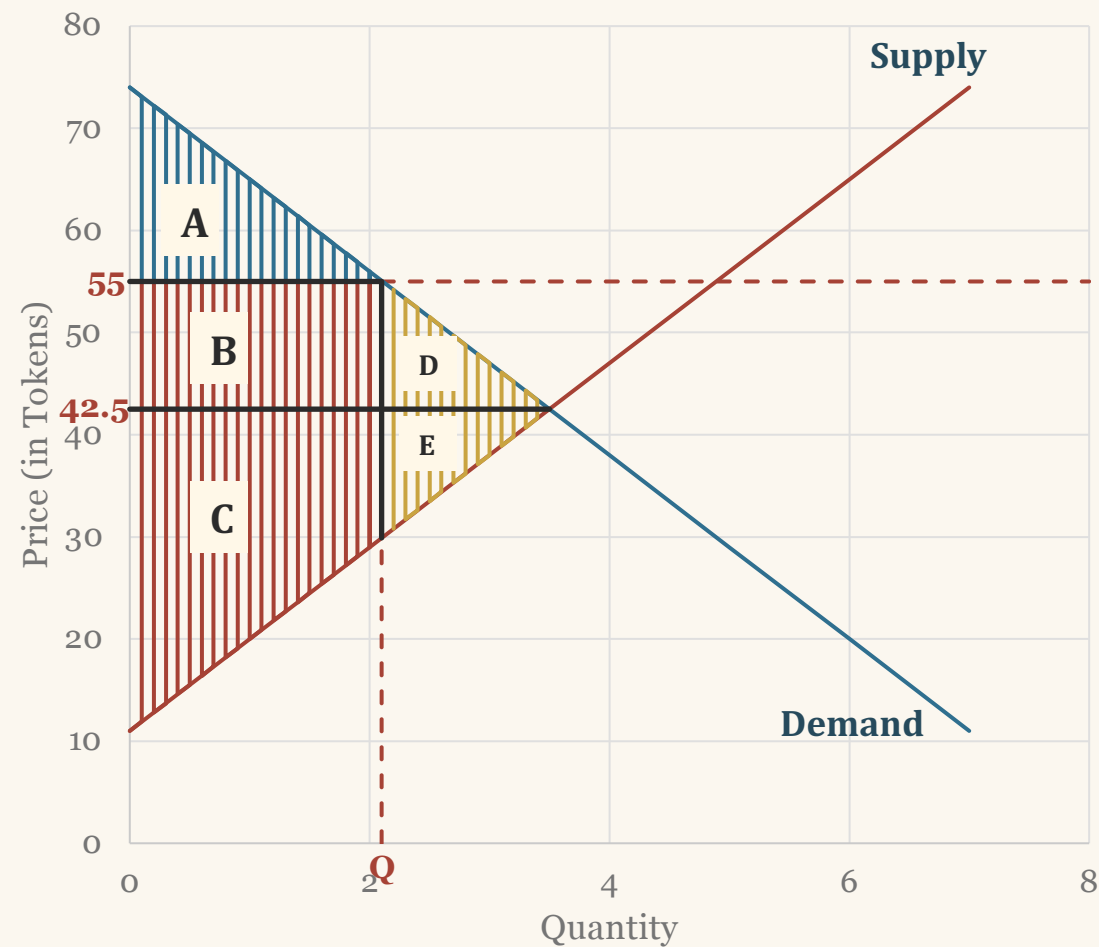
Mayor Intervenes



Deadweight Loss:

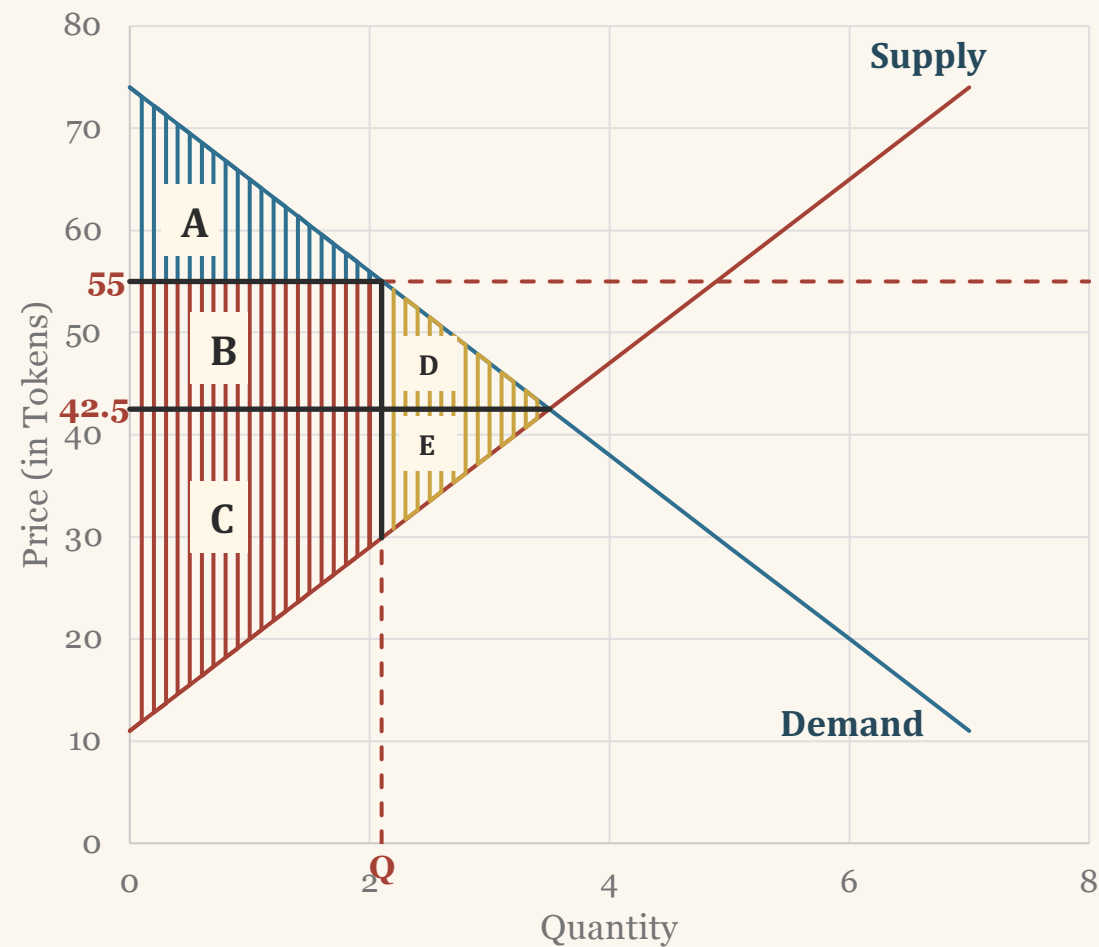
The reduction in economic surplus resulting from a market not being in competitive equilibrium is known as deadweight loss.

Mayor Intervenes



	At Market Equilibrium, P=42.5	At Price P=55
Consumer Surplus	A+B+D	A
Producer Surplus	C+E	B+C
Deadweight Loss	None	D+E

Mayor Intervenes



	At Market Equilibrium, P=42.5	At Price P=55
Consumer Surplus	A+B+D	A
Producer Surplus	C+E	B+C
Deadweight Loss	None	D+E

Government Interventions

Government can change markets by:

- Setting Prices
- Taxing Trades

Who gains? Who loses? What disappears?

Petition from Berry Holders

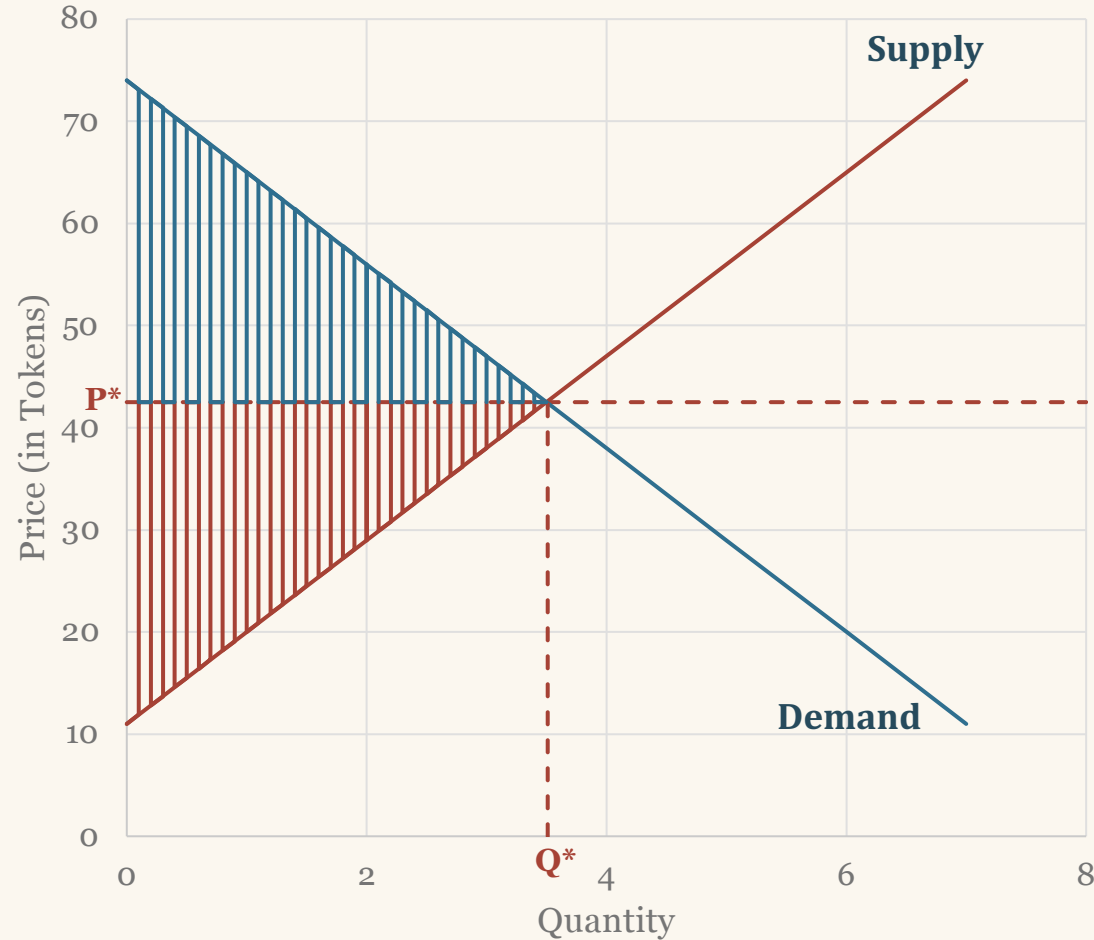
Citizens go to the Mayor:

“If berry prices stay too low, no one will store berries for next Founders’ Night.”



What shall the mayor do?

The Berry Support Price



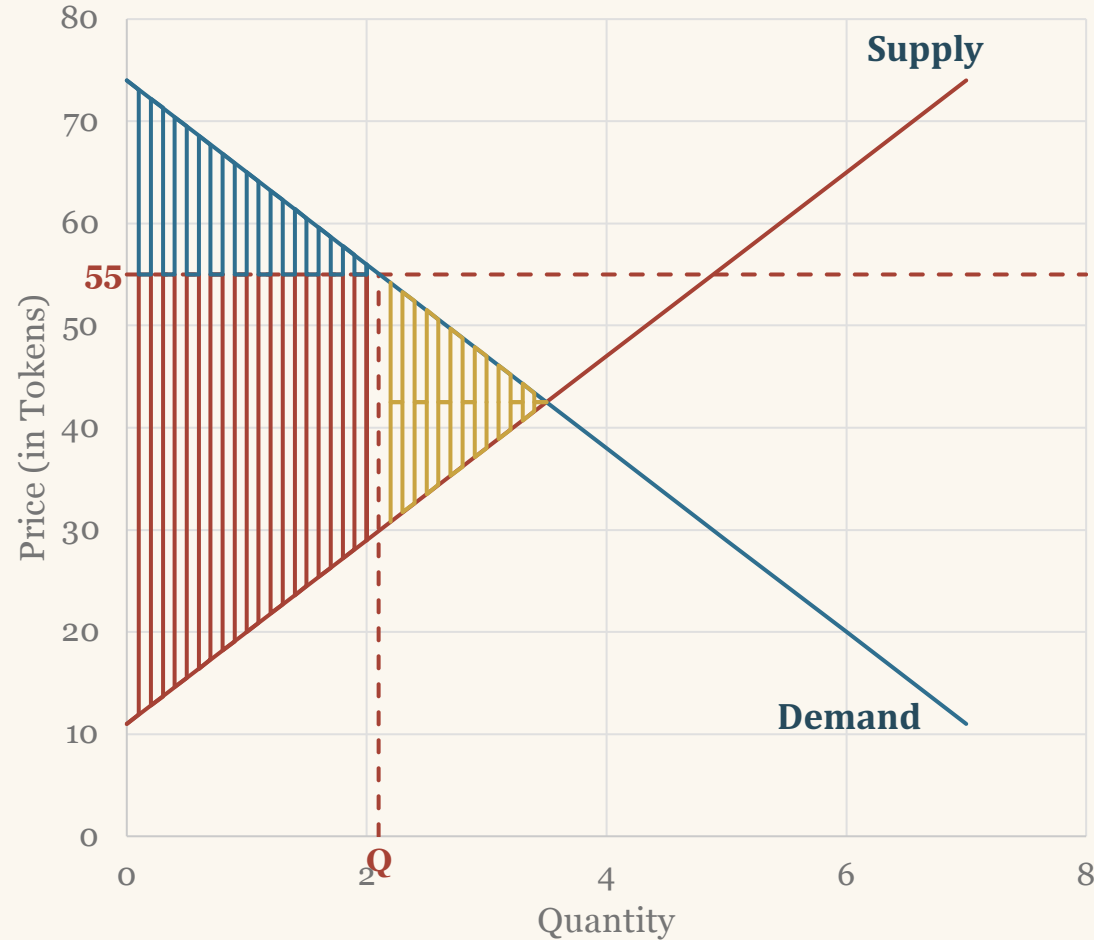
Mayor's rule:

No berry may sell below 55 Tokens.

This is a **price floor**:

A legally determined minimum price that sellers may receive.

The Berry Support Price



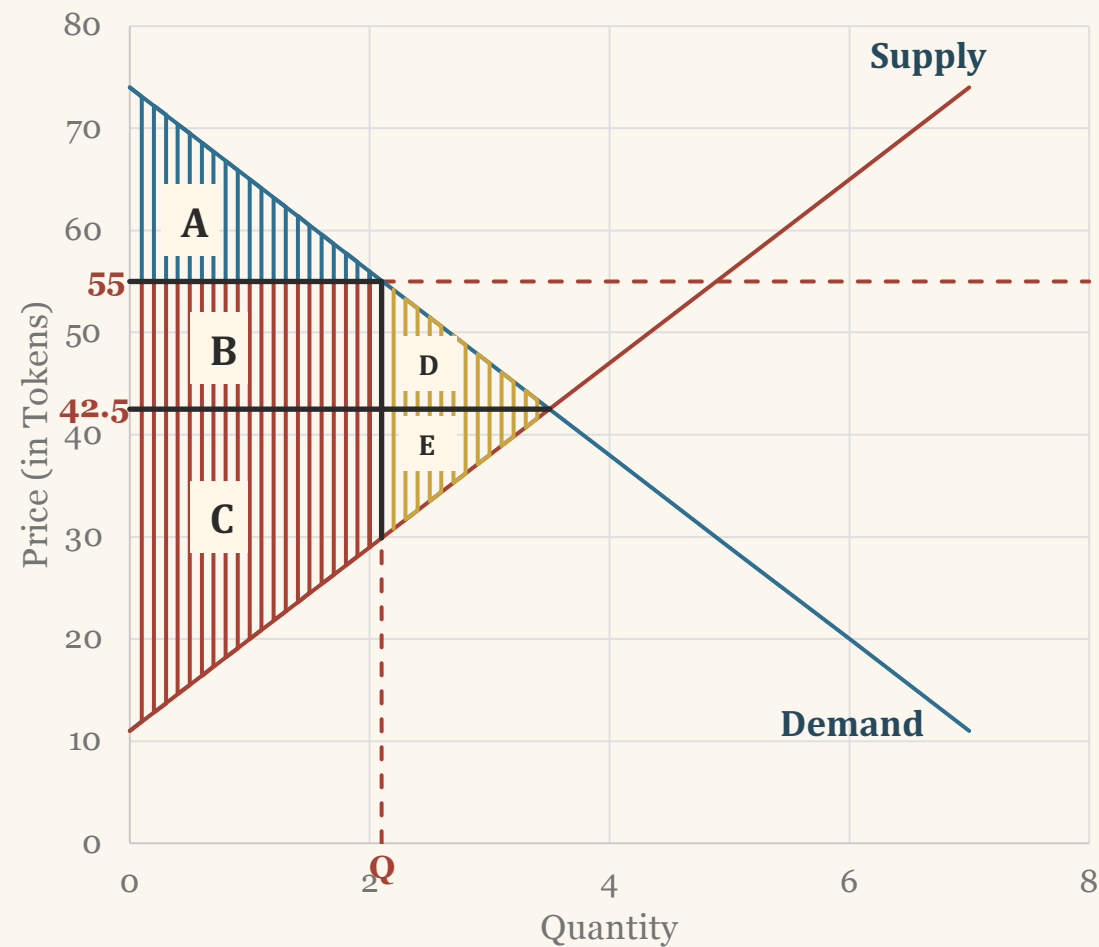
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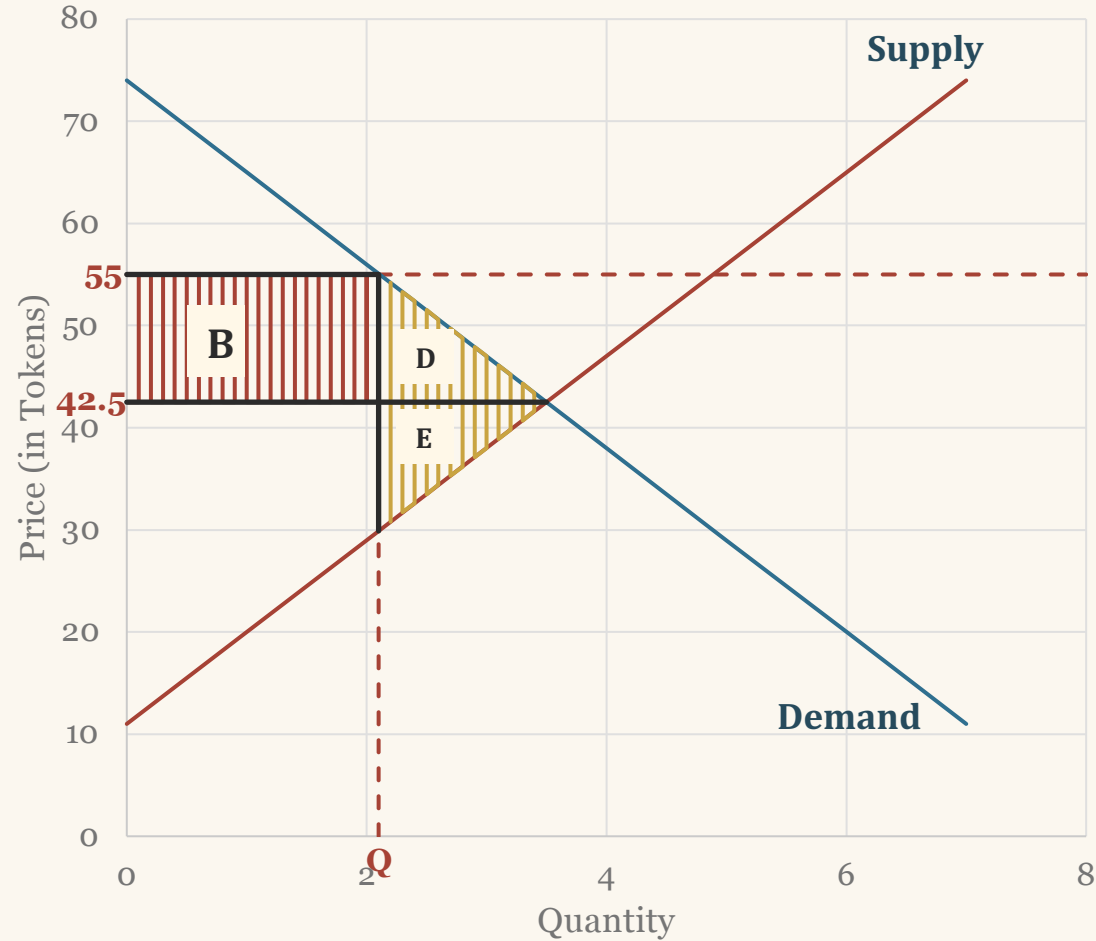
A legally determined minimum price that sellers may receive.

The Berry Support Price



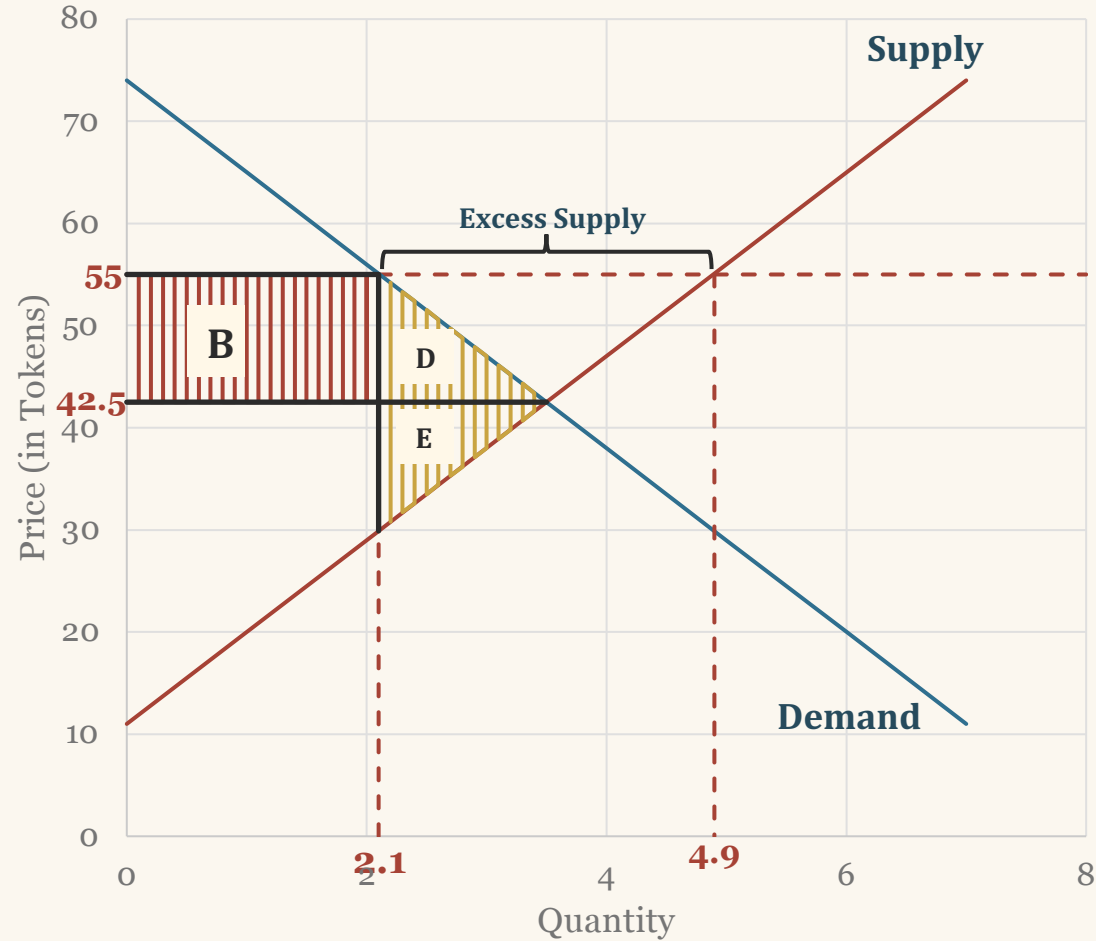
	At Market Equilibrium, $P=42.5$	At Price $P=55$
Consumer Surplus	$A+B+D$	A
Producer Surplus	$C+E$	$B+C$
Deadweight Loss	None	$D+E$

The Berry Support Price



D + E: Deadweight Loss
B: Consumer Surplus transferred to producers

The Berry Support Price



If the Berry Holders don't realize that they will not be able to store all berries:

Excess Supply

Petition from Pastry Consumers

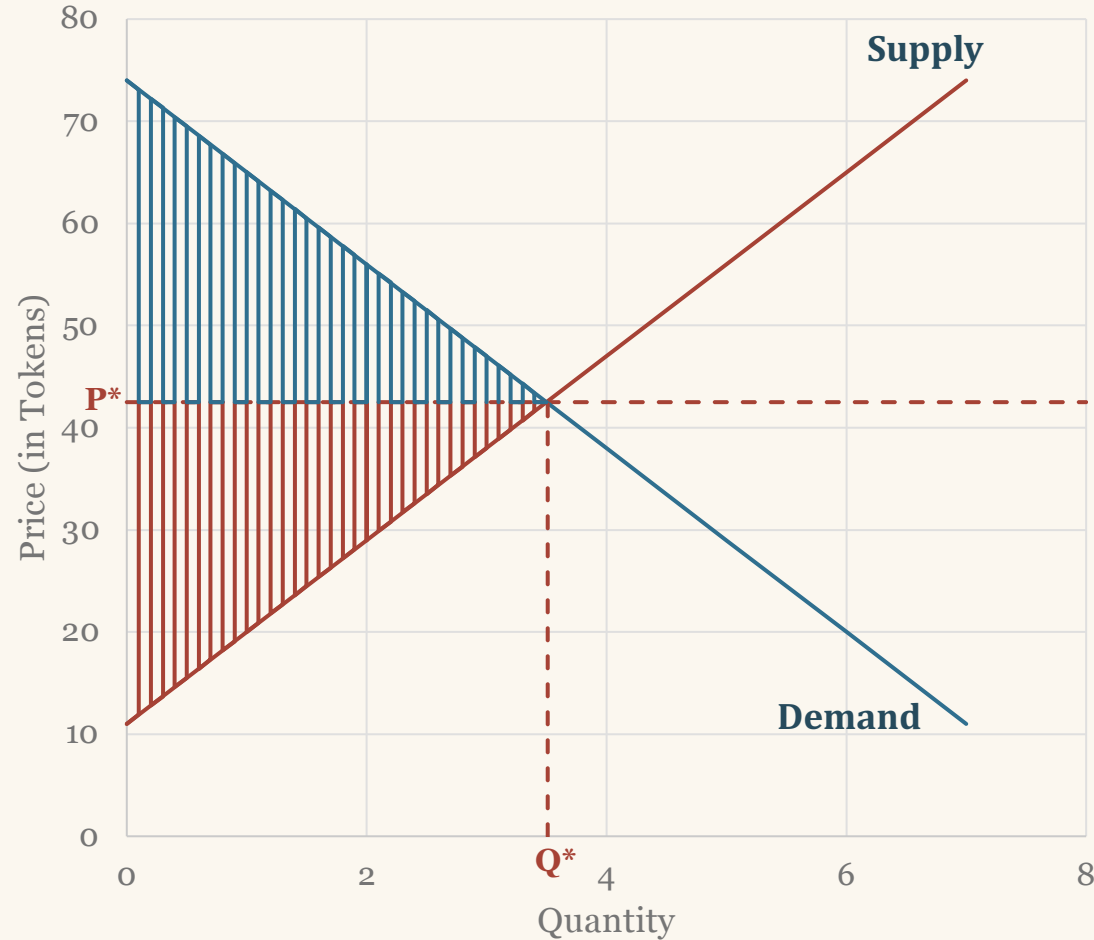
Citizens go to the Mayor:

*“We have fillings, but berries are too expensive.
Without berries, our pastries remain unfinished.”*



What shall the mayor do?

The Berry Price Cap



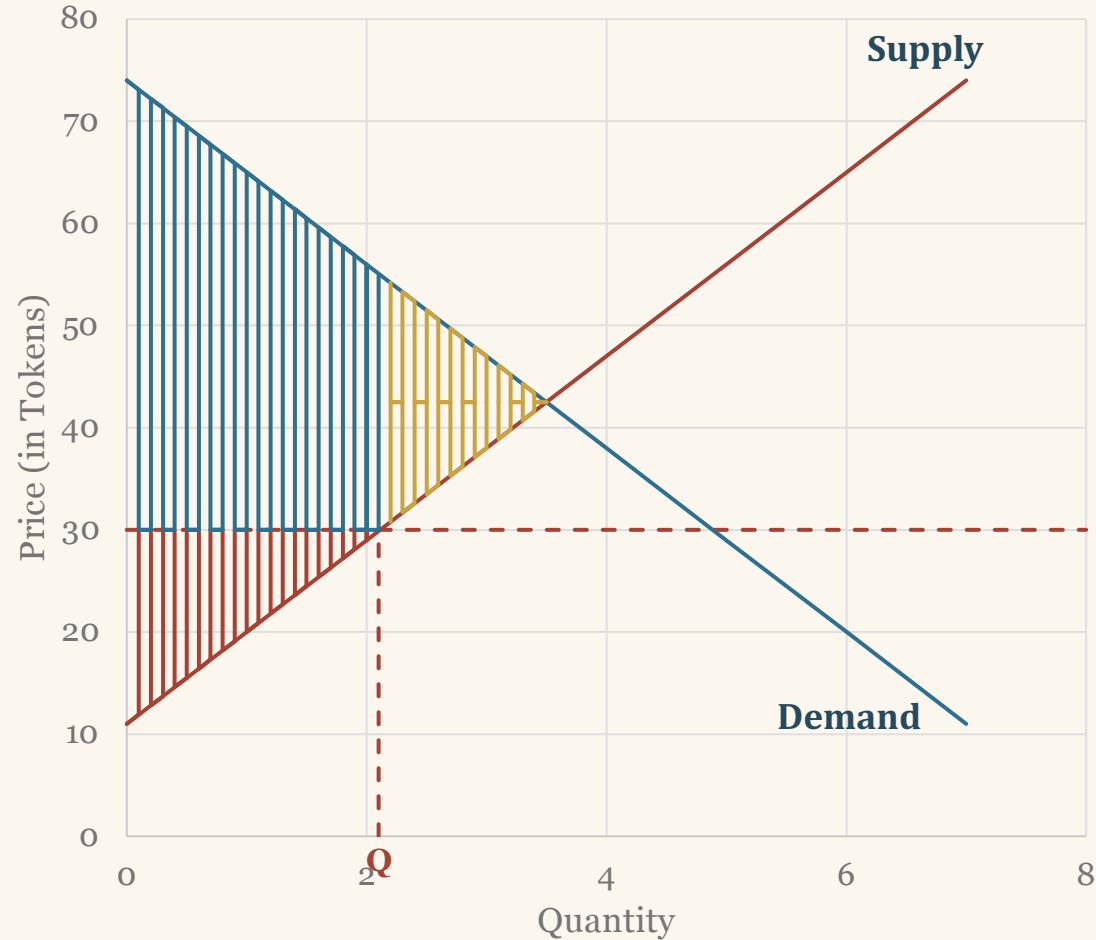
Mayor's rule:

No berry may sell above 30 Tokens.

This is a **rent/price ceiling**:

A legally determined maximum price that sellers may charge.

The Berry Price Cap



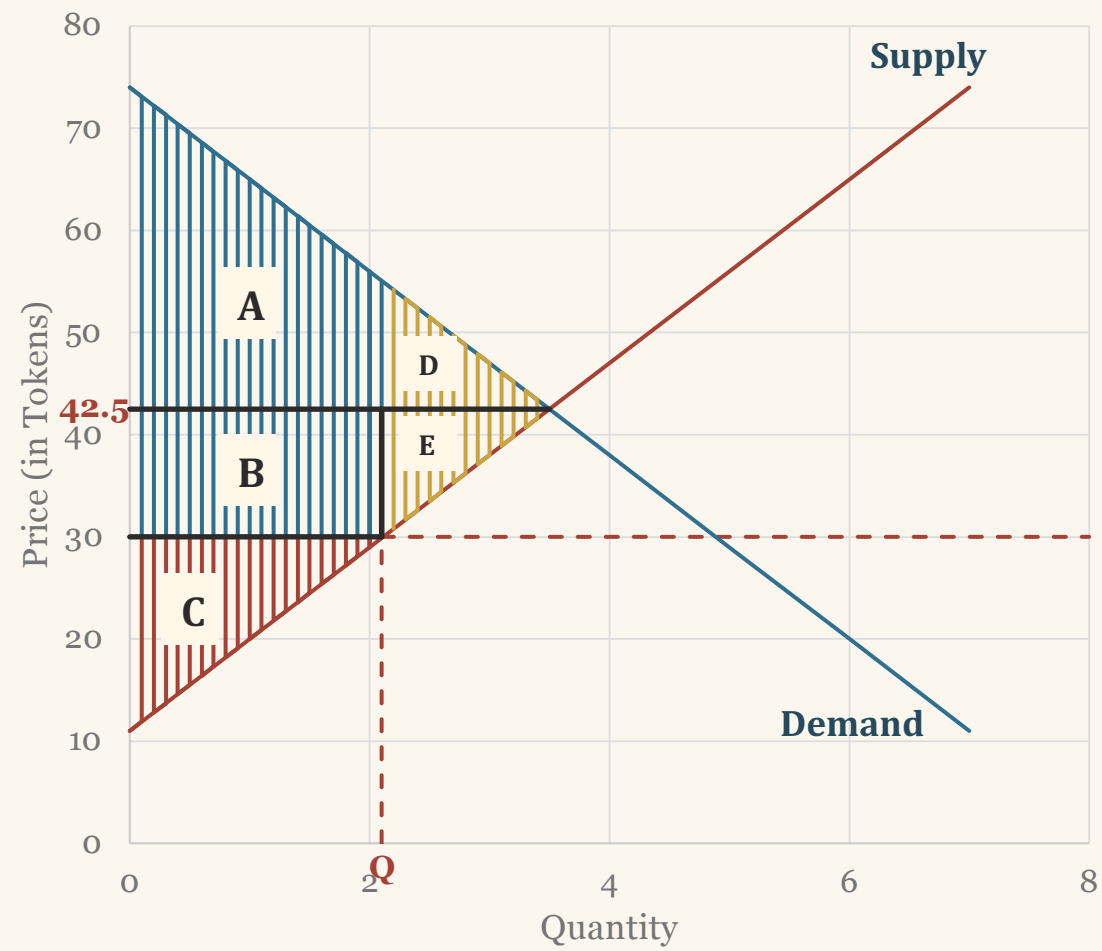
Mayor's rule:

No berry may sell above 30 Tokens.

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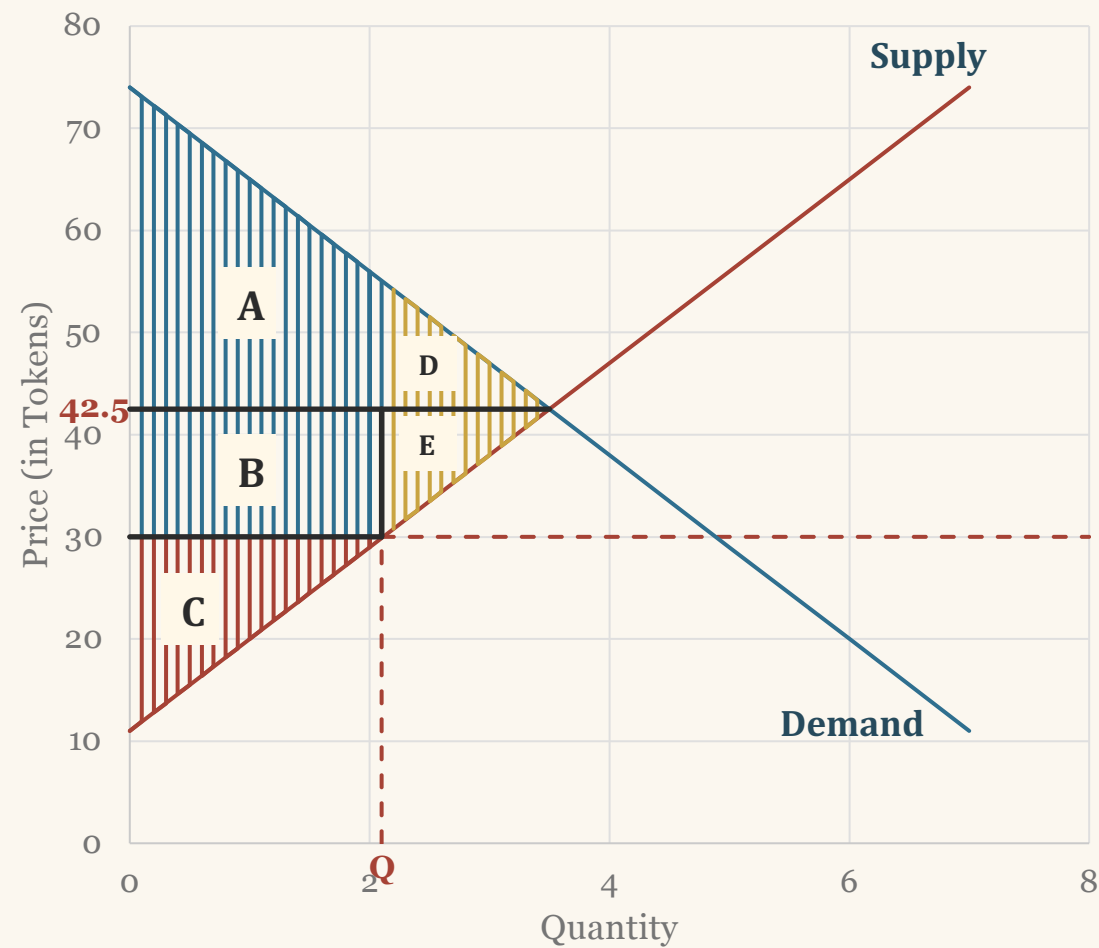
A legally determined maximum price that sellers may charge.

The Berry Price Cap



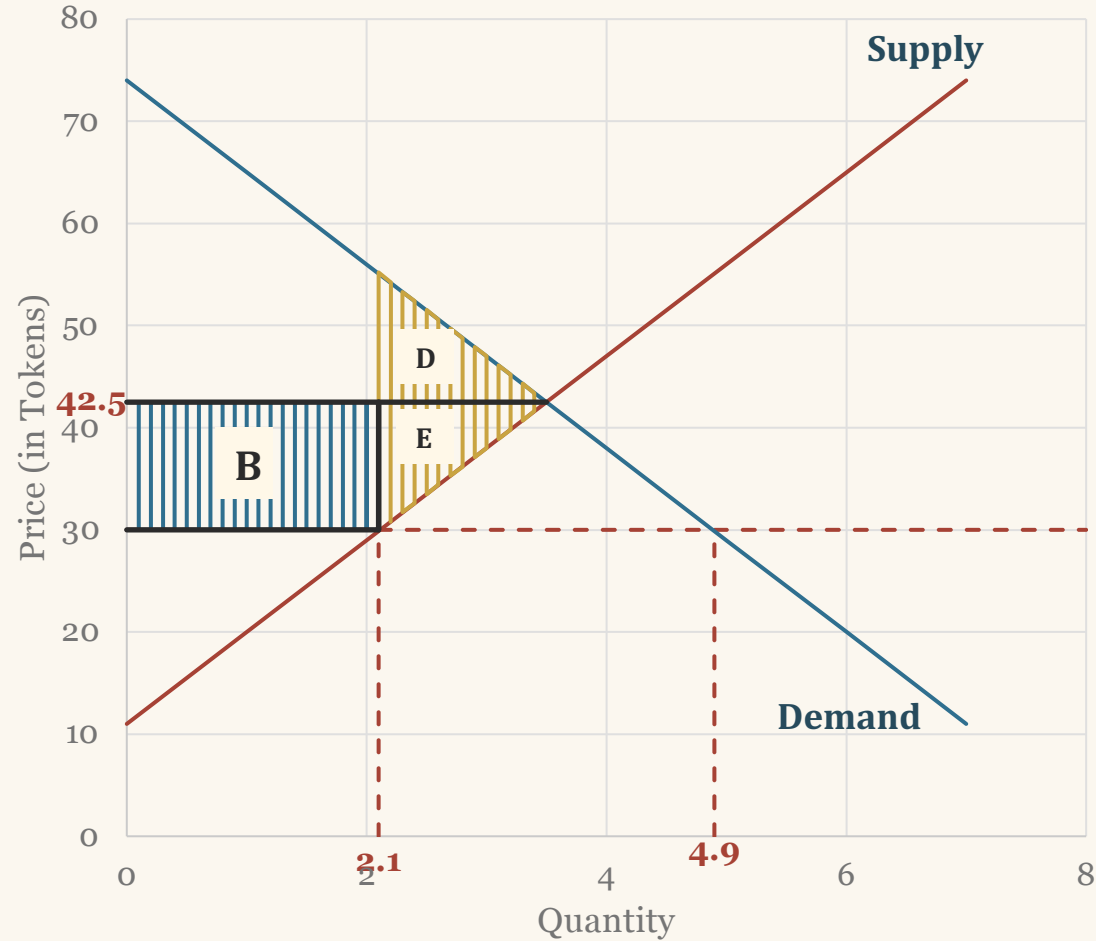
	At Market Equilibrium, $P=42.5$	At Price $P=30$
Consumer Surplus		
Producer Surplus		
Deadweight Loss		

The Berry Price Cap



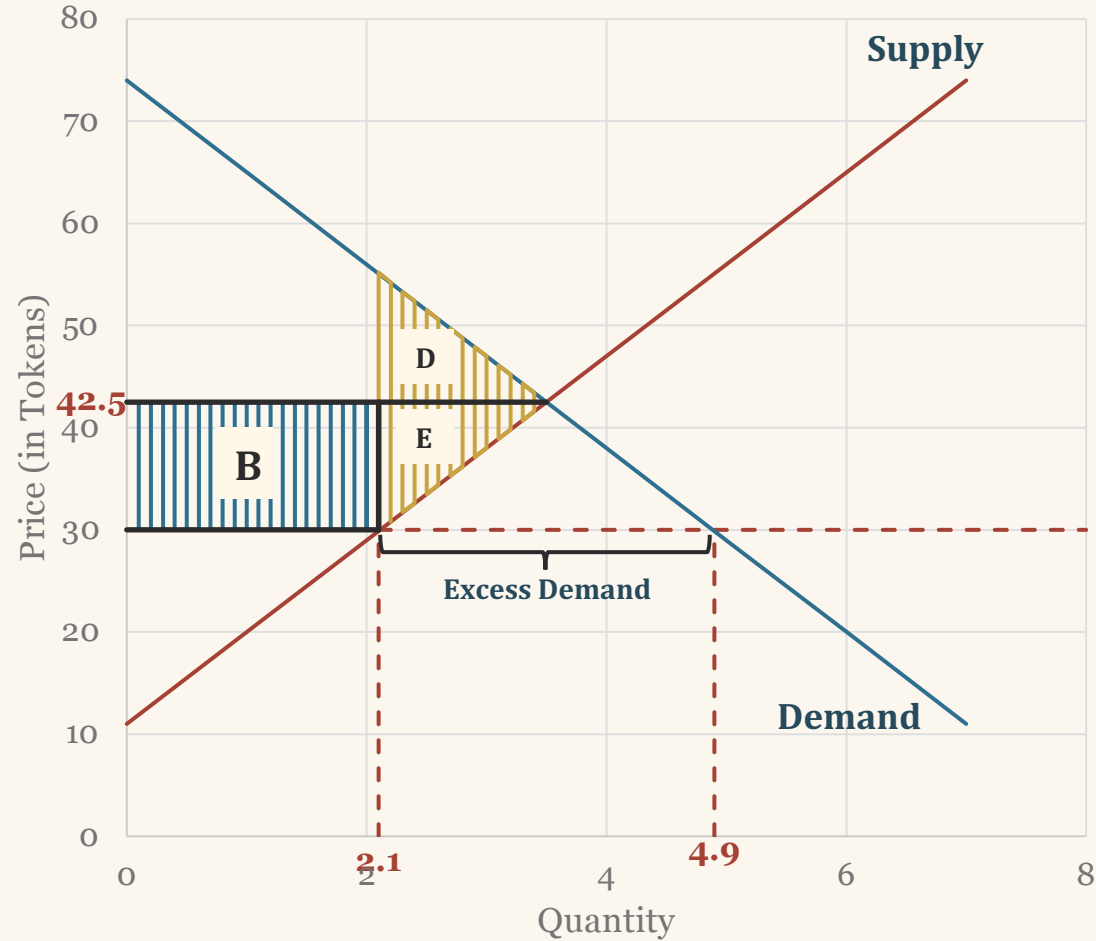
	At Market Equilibrium, $P=42.5$	At Price $P=30$
Consumer Surplus	$A+B+D$	$A+B$
Producer Surplus	$C+E$	C
Deadweight Loss	None	$D+E$

The Berry Price Cap



D + E: Deadweight Loss
B: Producer Surplus transferred to
Consumers

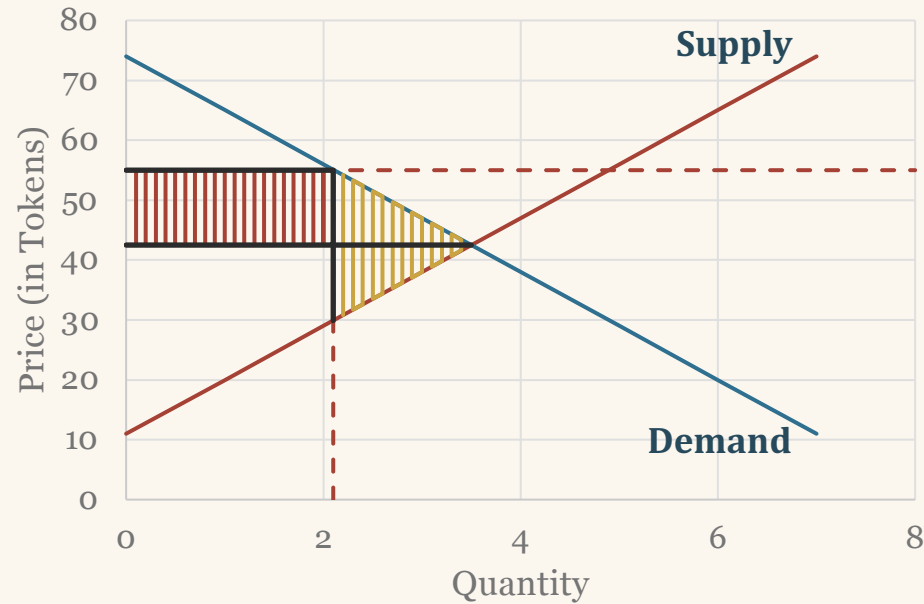
The Berry Price Cap



There is a shortage of berries in the Market:

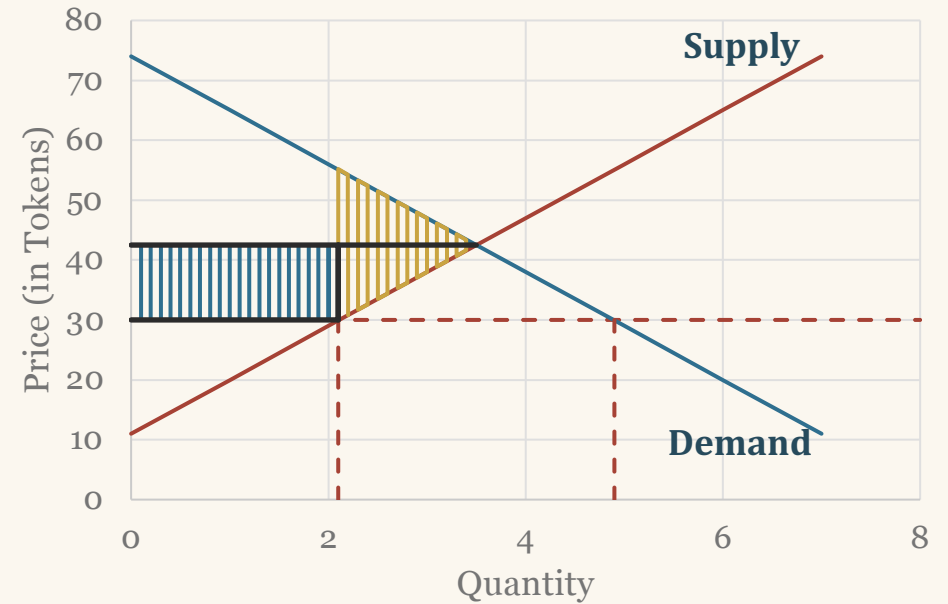
Excess Demand

Two Petitions, Two Outcomes



Price Floor

- Producers are made better off
- Consumers are made worse off
- Excess Supply



Price Ceiling

- Consumers are made better off
- Producers are made worse off
- Excess Demand

Price Setting in the US

- In the U.S., direct price setting is uncommon.
- Most prices are left to markets.
- But some examples exist:

Policy	Type	Market
Minimum wage	Price floor	Labor
Rent control / rent stabilization	Price ceiling	Housing
Agricultural price supports	Price floor	Farming

The Vault Repair Bill

After the petitions, the Town Treasurer arrives:

“The Berry Vault needs repairs before next Founders’ Night.”



Proposal: Every berry trade pays a small town-tax.

The Berry Stamp Tax

Town rule: *Every berry sale requires an 18-token stamp.*

This is a per-unit tax:

A payment to the government on each unit traded.

Example: The U. S. Federal government imposes an excise tax of 18.4 cents per gallon of gasoline as of 2026.

Effect of Tax

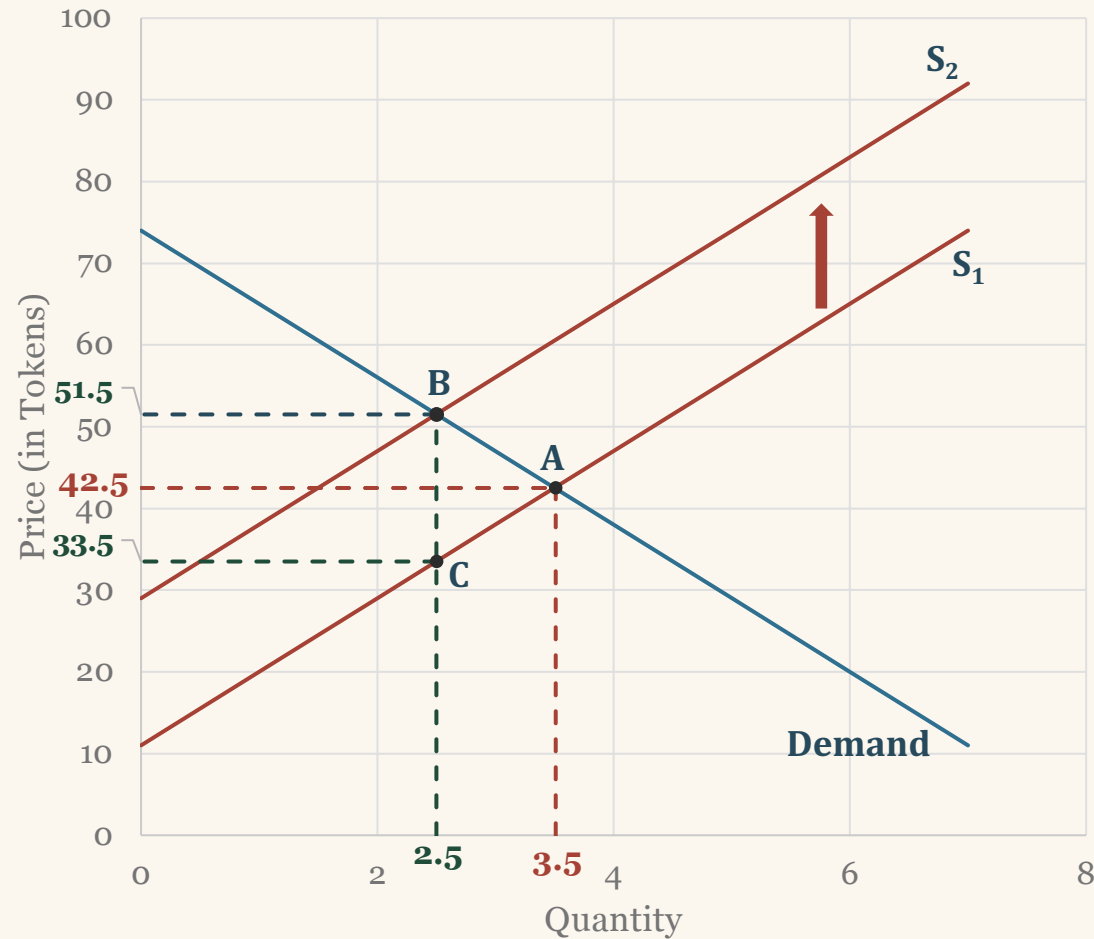


A tax of 18 tokens shifts the supply curve upwards.

Earlier: 3.5 berries at 42.5 Tokens

Now: 3.5 berries at 60.5 Tokens

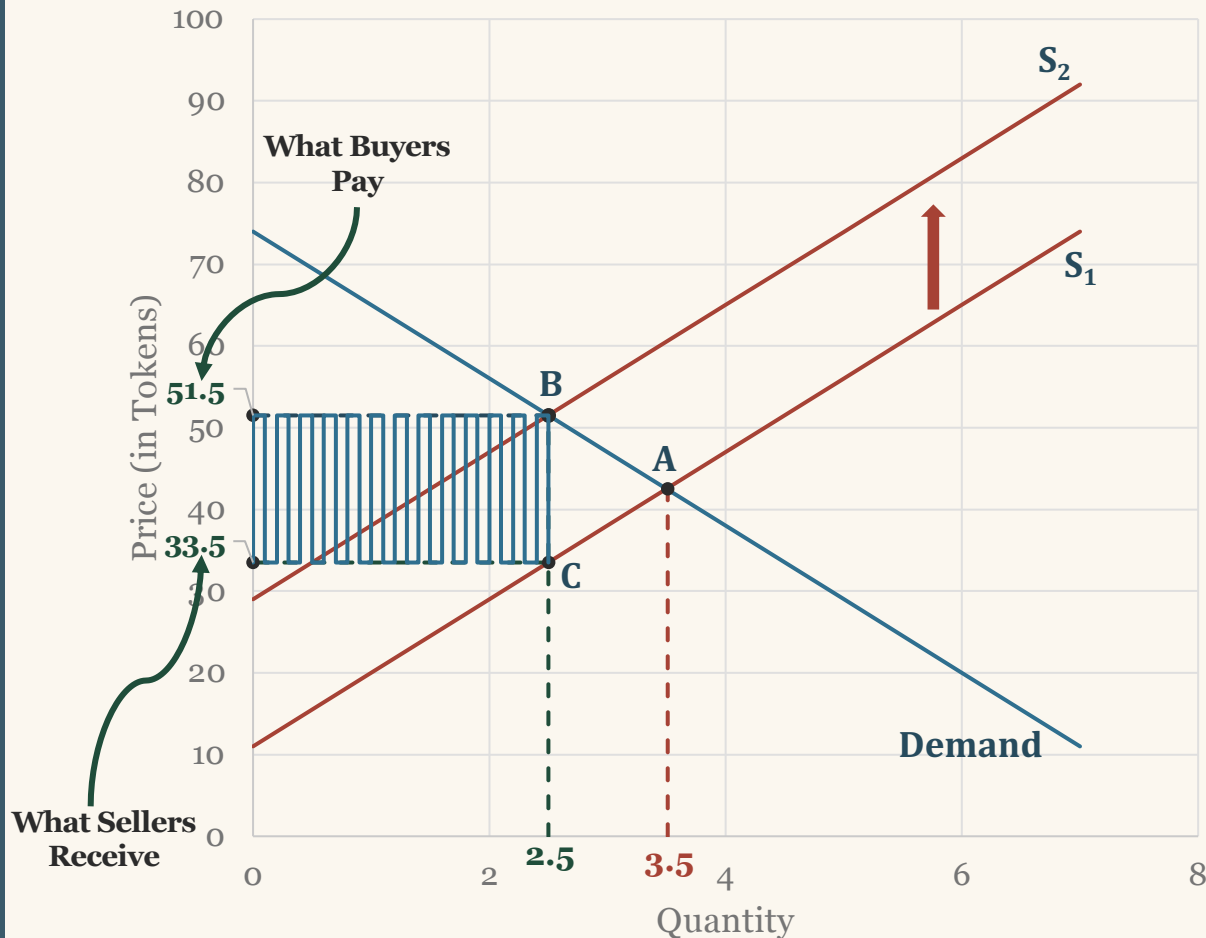
Effect of Tax



At New Equilibrium, B:
Berries sold at **51.5** → Buyers pay **51.5**

Sellers pay 18 Tokens in tax.
So, Sellers receive → **51.5 - 18 = 33.5**

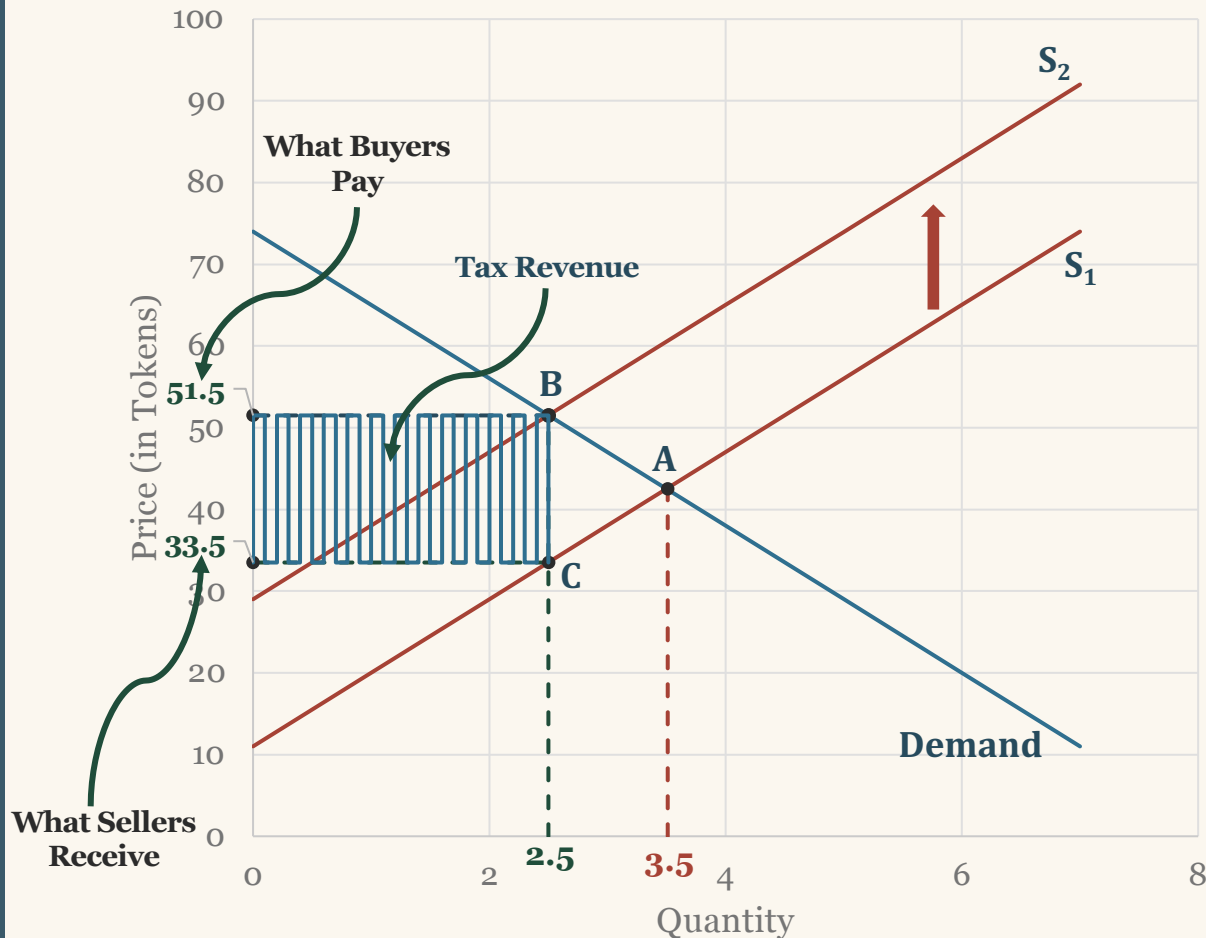
Effect of Tax



At New Equilibrium, B:
Quantity: 2.5

$$\text{Tax Revenue} = 18 \times 2.5 = 45$$

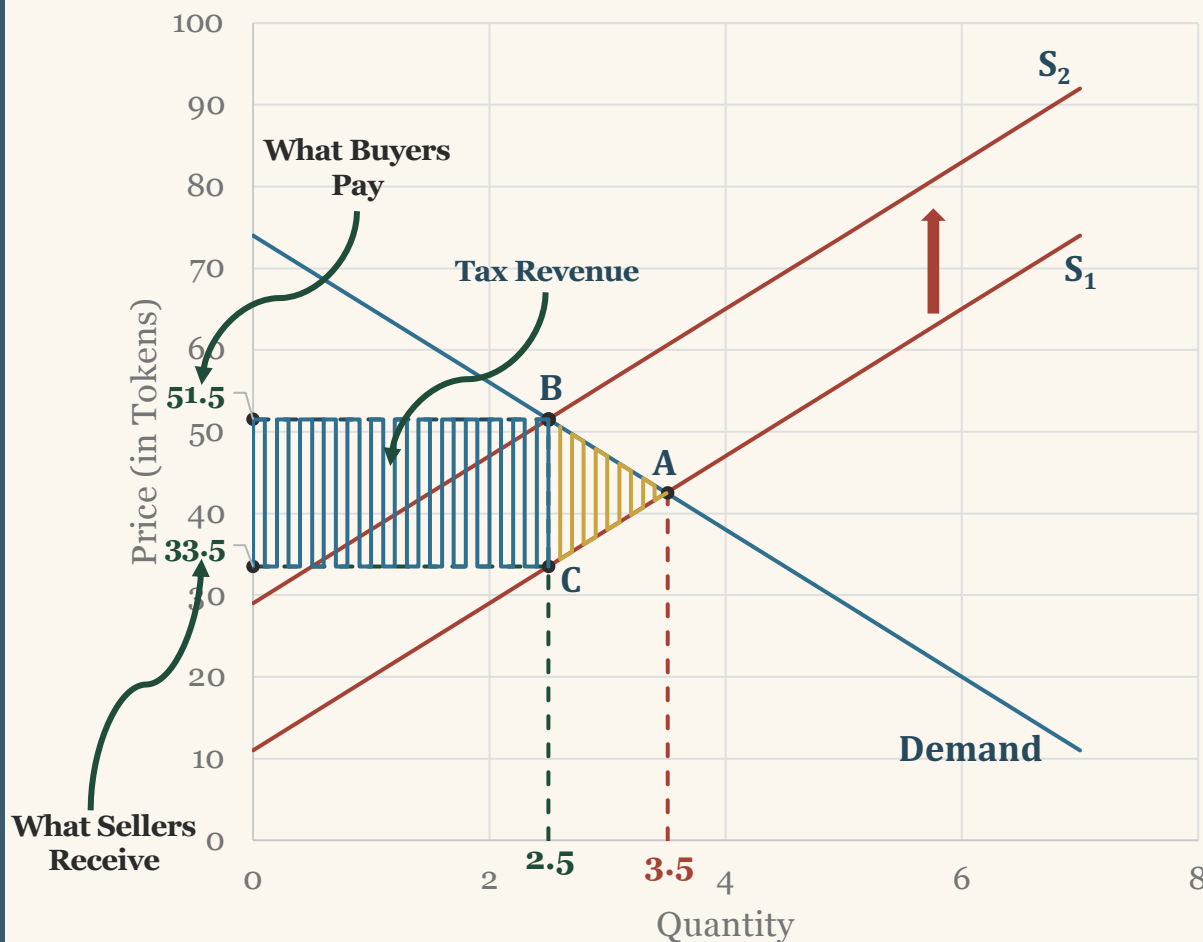
Effect of Tax



At New Equilibrium, B:
Quantity: 2.5

$$\text{Tax Revenue} = 18 \times 2.5 = 45$$

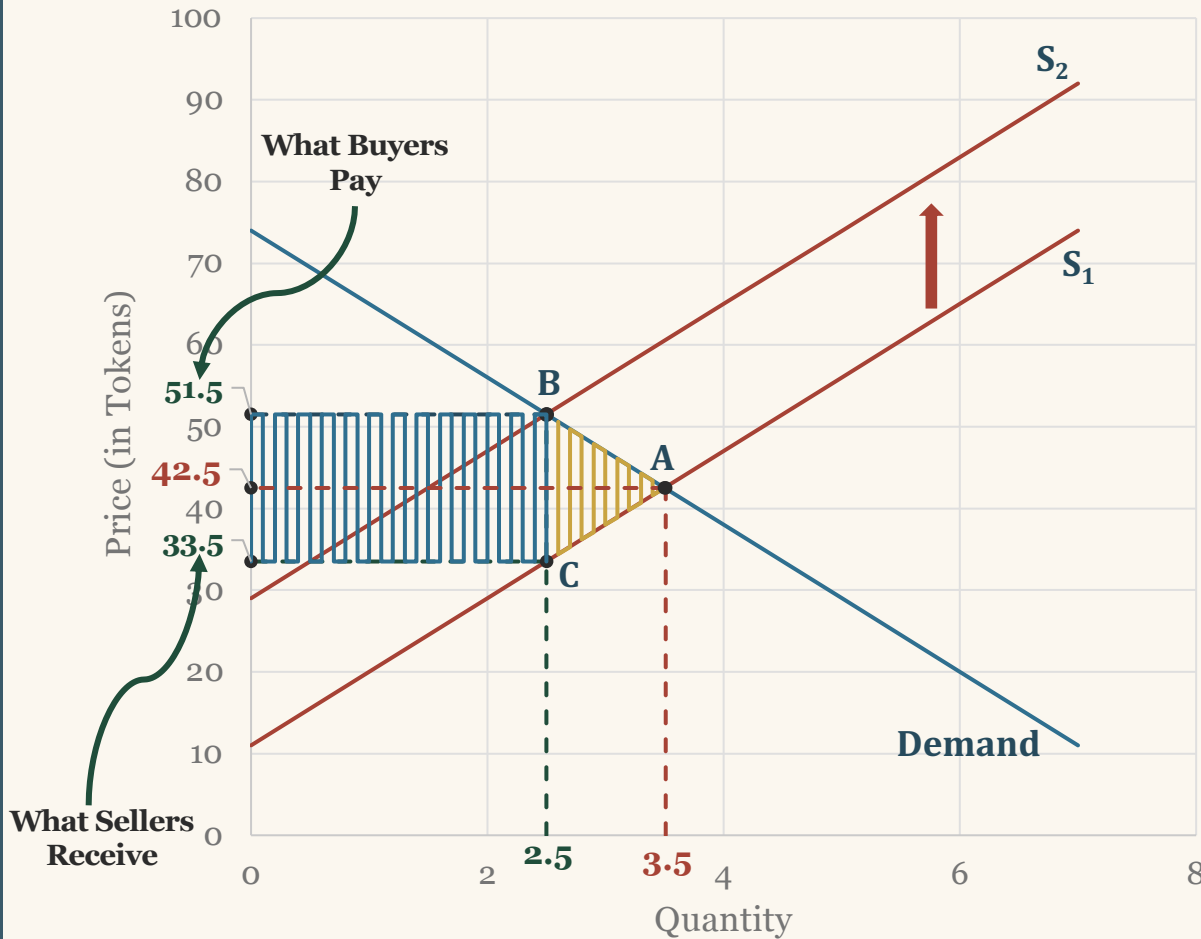
Effect of Tax



At New Equilibrium, B:
Quantity: 2.5

$$\text{Tax Revenue} = 18 \times 2.5 = 45$$
$$\text{Deadweight Loss} = \frac{1}{2} \times 18 \times 1 = 9$$

Who bears the Tax burden?



Old Price: 42.5

Buyers Pay: 51.5

Sellers receive: 33.5

Tax Burden on Buyers:

$$51.5 - 42.5 = 9$$

Tax Burden on Sellers:

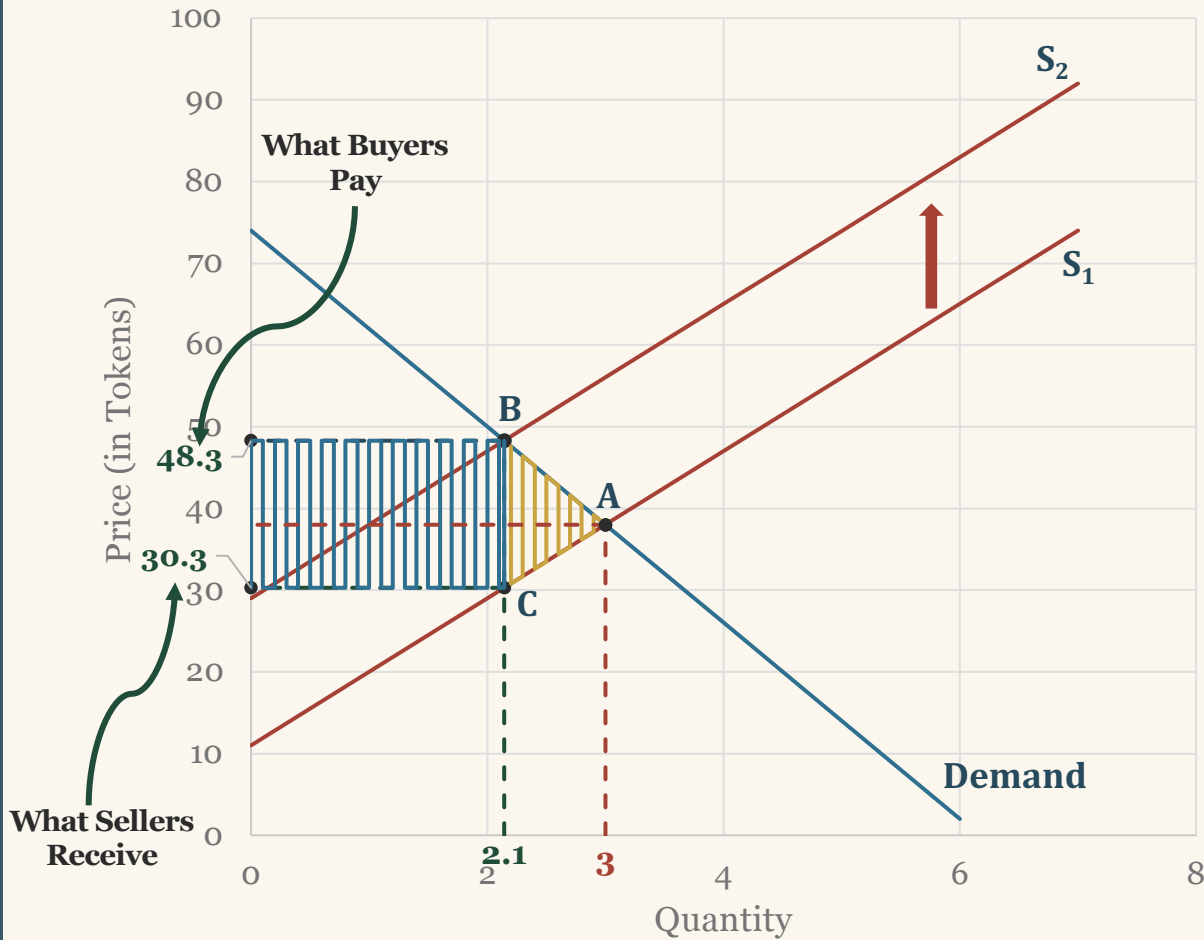
$$42.5 - 33.5 = 9$$

Who Bears more burden?

Tax burden depends on elasticity.

Less elastic side bears more of the tax.

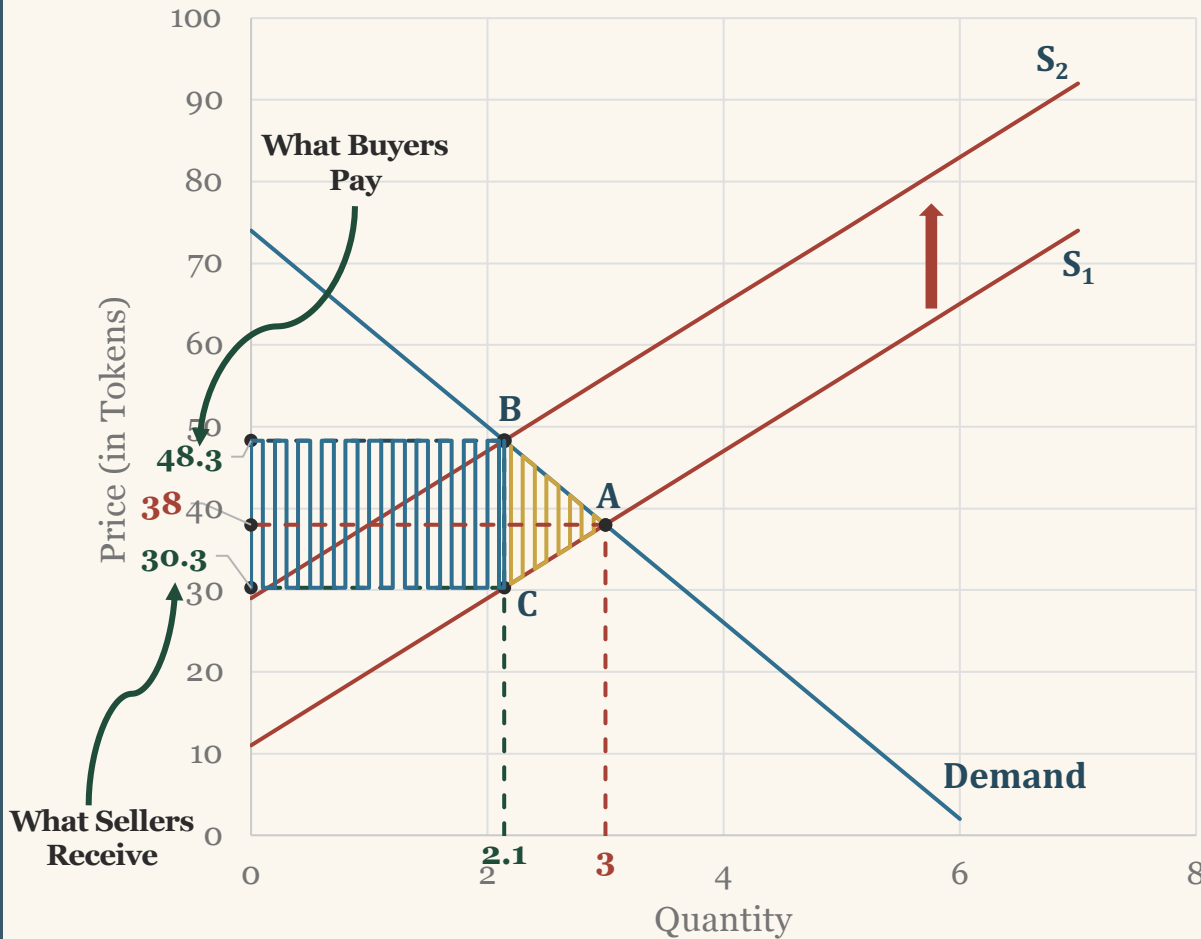
Who Bears more burden?



Tax burden depends on elasticity.

Less elastic side bears more of the tax.

Who Bears more burden?



Demand is less elastic than supply.



Consumers react less to an increase/decrease in price.



Consumers' burden of Tax:

$$48.3 - 38 = 10.3$$

Producers' burden of Tax:

$$38 - 30.3 = 8.7$$